

Switching of RNA splicing regulators in immature neuroblasts: a key step in adult neurogenesis

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Abstract

Summary

The lateral wall of the subventricular zone harbors neural stem cells (NSC, B cells) which generate proliferating transient-amplifying progenitors (TAP, C cells) that ultimately give rise to neuroblasts (NB, A cells). Molecular profiling at the single cell level struggles to distinguish these different cell types. Here, we combined transcriptome analyses of FACS-sorted cells and single-cell RNAseq to demonstrate the existence of an abundant, clonogenic and multipotent population of immature neuroblasts (iNB cells) at the transition between TAP and migrating NB (mNB). iNB are reversibly engaged in neuronal differentiation. Indeed, they keep molecular features of both undifferentiated progenitors, plasticity and unexpected regenerative properties. Strikingly, they undergo important progressive molecular switches, including changes in the expression of splicing regulators leading to their differentiation in mNB subdividing them into 2 subtypes, iNB1 and iNB2. Due to their plastic properties, iNB could represent a new target for regenerative therapy of brain damage.

eLife assessment

This **useful** manuscript presents an intriguing potential refinement of models for adult SVZ neurogenesis, and highlights the role of RNA splicing at specific stages in the lineage. Unfortunately, the evidence does not fully support the claims, leaving it currently **incomplete**. The proposed role of RNA splicing in neuronal differentiation, though interesting, remains unexplored and would benefit significantly from targeted gene manipulation studies.

Introduction

New neurons are constantly produced from neural stem cells (NSC) throughout life in two restricted neurogenic areas of the adult mammalian brain: the sub-granular zone (SGZ) of the hippocampus and the subventricular zone (SVZ) ¹. The lateral wall (LW) of the SVZ, adjacent to the striatum, is the largest germinal zone of the adult mouse brain, in which NSC (type B1 cells) are produced during the embryonic development and are mostly in a quiescent state thereafter^{2,3}. Upon activation, NSC divide and give rise to rapidly dividing transit-amplifying cells (TAP, type C cells), which ultimately produce neuroblasts (NB, type A cells). These neuroblasts migrate out of the SVZ anteriorly along blood vessels and the rostral migratory stream (RMS) to reach the olfactory bulb (OB) where they integrate and differentiate into various subtypes of interneurons required to fine-tune odor discrimination ¹. The diversity of the OB interneurons depends on the intrinsic regional identity of NSC in the SVZ along the anterior-posterior and medial-lateral axes^{1,4}.

Besides their neurogenic fate, NSC and neural progenitors present in the SVZ are also proficient to produce astrocytes and oligodendrocytes that migrate toward the corpus callosum, the RMS and the striatum in the normal^{5,6,7} and injured adult brain^{8,9}.

Transcriptomic analyses using different combinations of cell surface markers and transgenic mice models to isolate SVZ cells have provided important initial insights into stem cell quiescence and have highlighted consecutive transitions during neuronal lineage progression ^{2,10–13}. Since then, the single cell RNA-sequencing has revolutionized the field and has made it possible to precisely elucidate the transcriptome of SVZ cells present in the LW^{14–17} and in the septal wall which also harbours NSC niches¹⁸.

We have previously developed a FACS-based method, using a combination of three surface markers (LeX, EGF receptor, CD24), allowing to sort five distinct SVZ cell populations: quiescent NSC (LeX-EGFR-CD24-, hereafter referred to as sorted-qNSC, s-qNSC), activated NSC (LeX+EGFR+CD24-, referred to as sorted, s-aNSC), transit amplifying cells (LeX-EGFR+CD24-, referred to as sorted, s-TAP), immature (LeX-EGFR+CD24^{low}, referred to as sorted, s-iNB) and migrating neuroblasts (LeX-EGFR-CD24^{low}, referred to as sorted, s-mNB) from the adult mouse brain^{19,20}. Importantly, whereas s-iNB presented neuroblast markers, e.g. CD24 and doublecortin (DCX), s-iNB gathered the most abundant population among SVZ cycling progenitors (representing approximately 4.3% of total SVZ cells vs 1% s-aNSC and 2.5% s-TAP)²⁰.

We have previously shown that following a genotoxic stress (exposure to ionizing radiation) inducing the depletion of cycling progenitors, s-qNSC exit quiescence restoring successively s-aNSC, s-TAP, s-iNB and finally s-mNB²⁰.

Transcriptomic studies revealed that age-related SVZ neurogenesis decline is associated with TGF β signaling and specific cell cycle regulation changes in s-aNSC. A progressive lengthening of aNSC G1 phase with age leads to a decrease in the production of s-TAP, s-iNB and s-mNB ^{20–22}. However, despite the decrease of neurogenesis with aging the s-iNB remained more abundant than s-aNSC and s-TAP in the aged brain ²¹.

Studies of SVZ neural progenitors have been essentially focused on the characterization of the molecular mechanisms regulating quiescence, activation and the regenerative potential of aNSC. Here, we combined transcriptome analyses using DNA microarrays on FAC-sorted cells and single-cell RNAseq to demonstrate that iNB form a molecularly distinct class of SVZ cells, exhibiting molecular features of both neural progenitors and neuroblasts while keeping unexpected

regenerative capacity and plasticity. We identified important and sequential molecular switches occurring in these cells before their final differentiation in migrating neuroblasts. Altogether, these data led us to propose a revision of the current model of adult neurogenesis.

Results

Regenerative potential and multipotency of sorted iNB *in vitro* and *in vivo*

We have previously shown that s-iNB, which expressed the neuroblast markers CD24 and DCX²⁰ and markers of neural progenitors such as Mash1²⁰, Dlx2 (**Fig.S1A**) and SOX2 (**Fig.S1B**) had a clonogenic potential and were able to proliferate *in vitro* for weeks in normoxic conditions (20%O₂) unlike s-mNB²⁰. However, s-iNB exhibited reduced clonogenic potential compared to that of s-aNSC and s-TAP (**Fig.1A**). It has been previously reported that hypoxic conditions promoted SVZ NSC proliferation and neurogenesis^{23,24}. Hypoxic conditions (4%O₂) had no effects on s-mNB, but increased the proliferation capacities of s-aNSC, s-TAP and s-iNB, which reached similar growth rates, making them undistinguishable based on this endpoint (**Fig.1A**). Hypoxic conditions also increased the clonogenic potentials of both s-TAP and s-iNB but not that of s-aNSC (**Fig. 1B**). In these conditions, the clonogenic potential of s-TAP reached similar values as s-aNSC, but that of s-iNB was still lower (**Fig.1B**).

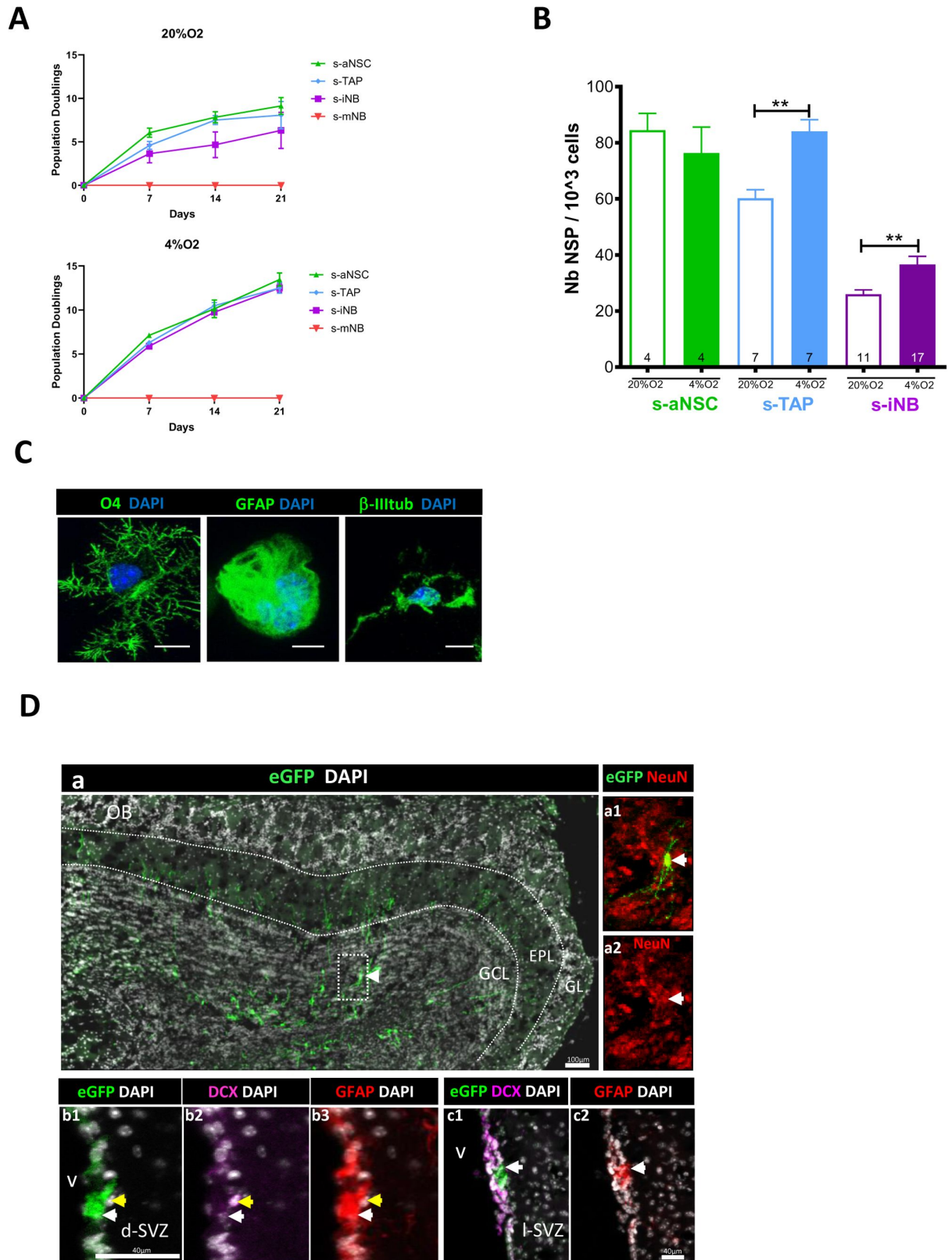


Figure 1

Plasticity of iNB *in vitro* and *in vivo*

s-aNSC, s-TAP, s-iNB, and s-mNB were sorted from the SVZ of 2-month old C57Bl/6 mice.

(A) Population doublings (PD). Data were obtained from 3 independent experiments (n=8).

(B) Clonogenicity assay. Results were obtained from 2 independent experiments. The number inside the bars indicated the number of microplate wells analysed (mean $\pm$ SEM).

(C) Differentiation assay. Representative images of immunofluorescence of freshly sorted iNB cells cultured in oligodendrocyte, astrocyte, or neuronal differentiation medium, and stained for NG2 and CNPase, GFAP and CD133 or β III-Tubulin and doublecortin (DCX) expressions, respectively. *p<0.05, **p<0.01, ***p<0.001. Scale bar: 20 μ m.

(D) EGFP-positive s-iNB were isolated from β -actin:eGFP mice and transplanted unilaterally at 3 injection points at proximity of the dSVZ/RMS of recipient C57Bl/6 mice. Transplanted brains were analysed five weeks later by immunostaining. **(a)** Detection of eGFP⁺ cells in the granule cell layer and the external cell layer of the olfactory bulb of a mouse transplanted. **(a1-a2)** a2 high magnification of the inset (dotted line in a) showing eGFP⁺ NeuN⁺ cells (white arrows). **(b-c)** Detection of eGFP⁺ GFAP⁺ DCX⁺ cells (yellow arrows) and eGFP⁺ GFAP⁺ (white arrows) in the dorsal (b2) and lateral (c2) SVZ. GCL:granule cell layer, EPL : external plexiform layer, GL: glomerular layer, ISVZ: lateral SVZ, dSVZ: dorsal SVZ, Scale bars= 40 μ m or 100 μ m.

Quite surprisingly, freshly isolated s-iNB had the capacity to generate the three neural lineages, neurons, astrocytes and oligodendrocytes, when plated for 5–7 days in the appropriate differentiation media, similarly as s-aNSC and s-TAP ([Fig.1C](#)).

Thus, to assess the regenerative potential of these immature neuroblasts *in vivo*, eGFP⁺ iNB were sorted from β -actin:eGFP mice, transplanted into the left striatum of recipient C57Bl/6 mice and compared to eGFP⁺s-mNB and eGFP⁺s-NSC/TAP. Five weeks after transplantation, eGFP⁺ cells were detected in different regions of interest ([Table 1](#)) in the brains of mice transplanted with eGFP⁺s-NSC/TAP or eGFP⁺s-iNB, whereas no eGFP⁺ were observed after transplantation of eGFP⁺s-mNB even at the injection site (IS) ([Table 1](#)). Excepted in one out of 3, animals transplanted with eGFP⁺s-iNB presented high levels of eGFP⁺ cells in the OB with levels equivalent or superior to animals transplanted with eGFP⁺s-NSC/TAP or eGFP⁺s-iNB ([Fig. 1D](#)). EGFP⁺ cells were detected in the rostral migratory stream ([Fig. S1A](#)) as well as in the granular cell layer (GCL) and the external plexiform layer (EPL) of the OB where they differentiated into NeuN⁺ neurons in 2/3 animals transplanted with eGFP⁺s-iNB ([Fig.1Da](#)) and in 3/3 animals transplanted with EGFP⁺s-NSC/TAP ([Table 1](#)).

Importantly, eGFP⁺ cells were present in the SVZ of all the animals transplanted with eGFP⁺s-iNB and eGFP⁺s-NSC/TAP ([Fig. 1Db](#), [Fig. 1Dc](#)), some of them expressing GFAP indicating the generation of astrocytes, and therefore possibly NSC ([Fig. 1Db3](#), [Fig. 1Dc2](#)). Plasticity of s-iNB was also shown at the IS by the detection of eGFP⁺ cells expressing markers of astrocytes (GFAP) and oligodendrocytes (CNPase) ([Fig.S2B](#) and [C](#)).

Overall, these results demonstrate the plasticity of s-iNB *in vitro*, making them very similar to s-NSC/TAP, questioning the exact nature of this abundant SVZ cell populations that express neuroblast markers.

		IS		SVZ		CC		OB		Total eGFP+ cells
		Nb eGFP+ cells	%	Nb eGFP+ cells	%	Nb eGFP+ cells	%	Nb eGFP+ cells	%	
s-aNSC/s-TAP 1.3-1.5x10 ⁴ eGFP+ engrafted cells	Animal 1	271	62.6%	17	3.9%	55	12.7%	90	20.8%	433
	Animal 2	300	31.9%	79	8.4%	47	5.0%	515	54.7%	941
	Animal 3	95	37.1%	73	28.5%	22	8.6%	66	25.8%	256
s-iNB 2x10 ⁴ eGFP+ engrafted cells	Animal 1	1025	65.2%	16	1.0%	49	3.1%	483	30.7%	1573
	Animal 2	196	15.5%	41	3.2%	41	3.2%	989	78.1%	1267
	Animal 3	240	84.8%	43	15.2%	0	0.0%	0	0.0%	283
s-mNB 2x10 ⁴ eGFP+ engrafted cells	Animal 1	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0
	Animal 2	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0
	Animal 3	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0

IS = Injection Site
CC = Corpus Callosum
SVZ = Subventricular zone
OB = Olfactory Bulb

Table 1

Transplantations of eGFP⁺s-iNB, eGFP⁻s-iNB and eGFP⁺s-mNB freshly isolated from β -actin:eGFP mice model

Table recapitulating the immunohistological analyses of the numbers of eGFP⁺ cells in different regions in recipient C57Bl6/J mice brains, 5 weeks after transplantation.

s-iNB are a molecularly distinct class of proliferating SVZ progenitors

We characterized the transcriptomic profiles of each SVZ populations sorted from 2 month-old C57Bl/6 mice using Clariom™ D mouse microarrays that provides intricate transcriptome-wide gene- and exon-level expression profiles (**Fig. 2A** [↗](#)).

Pseudotime analysis (TSCAN) [25](#) [↗](#) based on 3679 highly dispersed genes between the five sorted cell populations confirmed the lineage progression from s-qNSC (cluster1), then s-aNSC and s-TAP within a single cluster (cluster2), followed by s-iNB (cluster3) and finally s-mNB (cluster4) (**Fig. 2B** [↗](#)). Another pseudotime analysis excluding s-qNSC, based on 1481 highly dispersed genes, delineated two distinct clusters for s-aNSC and s-TAP (**Fig. 2C** [↗](#)), highlighting their molecular differences.

Venn Diagram comparison of data sets revealed 2950, 315, 129, 395 and 1693 genes unique to s-qNSC, s-aNSC, s-TAP, s-iNB and s-mNB respectively (Supp_data_1). G:Profiler functional profiling of these transcriptome signatures with KEGG and reactome databases uncovered significant stage-dependent changes (**Fig. S3** [↗](#), Supp_data_1). S-aNSC overexpressed genes involved in metabolism of glyoxylate and dicarboxylate metabolism. S-qNSC rather overexpressed genes related to metabolism of lipids and Rho GTPase cycle. S-TAP significantly displayed genes involved in Mitotic, G1 phase and G1/S transition, Activation of the pre-replicative complex. S-mNB specifically overexpressed genes involved in axon guidance and nervous system development. Conversely, s-iNB signatures contained many genes linked to cell cycle, prometaphase, M phase, and cell cycle checkpoints, clearly distinguishing them from the other cell populations including s-mNB.

Alternative splicing (AS) plays various functions in brain development including neural cell differentiation and neuronal migration [26](#) [↗](#)–[30](#) [↗](#). Using the splicing index (SI) corresponding to a value assigned to each exon by estimating its abundance with respect to adjacent exons using TAC software (ThermoFisher Scientific), we identified differentially spliced genes (DSG, adjusted p-value ≤ 0.05) unique to s-qNSC, s-aNSC, s-TAP, s-iNB and s-mNB (Supp_data_2). GO annotations of DSG (**Fig. S4** [↗](#)) clearly revealed that spliced genes in SVZ cells are mainly involved in neuron development and neurogenesis. Interestingly this also showed that qNSC logically differed from the other cell types by splicing concerning genes involved in mitosis and cell cycle, consistently with their quiescent state. More importantly, GO annotations of DSG confirmed that s-TAP and s-iNB have distinct features.

We further analysed particularly DSG at the successive cell transitions along neurogenesis. Results show a very large number of DSG between s-qNSC vs s-aNSC (9484) then no DSG at the s-aNSC and s-TAP transition, only 169 between the s-TAP and s-iNB transition, and finally 3759 between s-iNB and s-mNB (Supp_data_3). Functional profiling analysis with g:Profiler linked the DSG between s-TAP and s-iNB to the nervous system development and those between s-iNB vs s-mNB to protein binding or cell junction.

We then focused on the differentially expressed genes (adjusted p-value ≤ 0.05) belonging to the nine families of RNA splicing regulators (RSR) that play essential roles in the neurogenic pathway (*Ptbp*, *Nova*, *Rbfox*, *Elavl*, *Celf*, *Dbhs*, *Msi*, *Pcbp*, and *Mbnl*) [26](#) [↗](#). Strikingly, a clear down-regulation of most RSR gene expression distinguishes s-qNSCs from the other populations (Supp_data_4). Interestingly another shift discriminates s-aNSC and s-TAP from s-mNB (Supp_data_4). This led us to identify several groups of RSR genes in function of their variation of expression along with neurogenesis: the groups 1 and 6 were overexpressed in s-aNSC and s-TAP and downregulated in s-mNB; the group 2, 3 and 5 were up-regulated in mNB as compared to s-aNSC and s-TAP. All these groups were up-regulated in s-iNB, suggesting that the switch in RSR expression between cycling

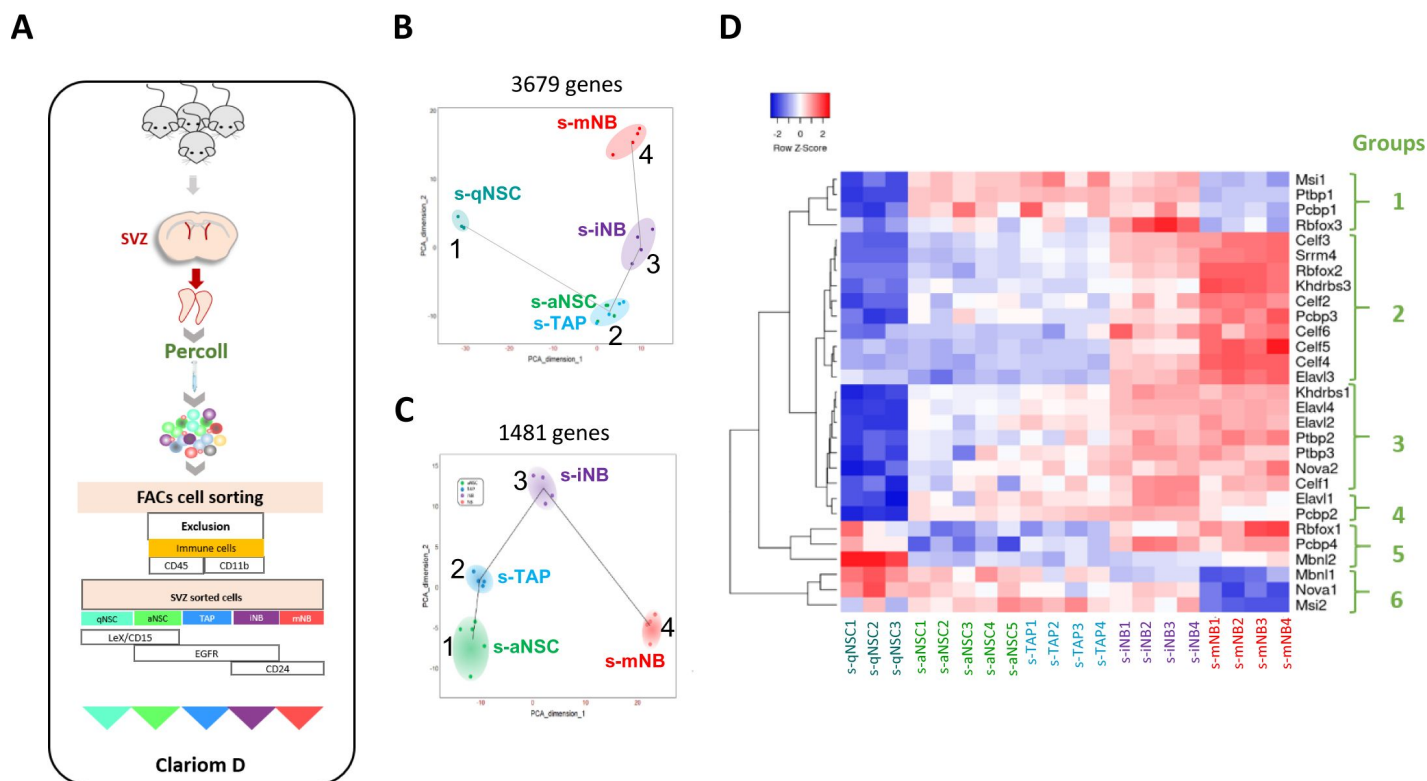


Figure 2

Transcriptomic analysis of SVZ neural progenitor cells

(A) Schematic illustration of our strategy of flow cytometry cell sorting of the five major neurogenic populations (s-qNSC, s-aNSC, s-TAP, s-iNB, s-mNB) of SVZ microdissected from adult mouse brains. **(B)** Pseudotime analysis showing the lineage progression of sorted cell populations based on the highly dispersed genes between s-qNSC, s-aNSC, s-TAP, s-iNB and s-mNB. **(C)** Pseudotime analysis after exclusion of s-qNSC based to further discriminate s-aNSC and s-TAP. **(E)** Heatmap representation of RSR genes differentially expressed among s-aNSC, s-TAP, s-iNB, s-mNB (FDR p-value < 0.05).

progenitors and s-mNB occurred in these cells (**Fig. 2D**). Interestingly, *Ptbp1*, encoding a major splicing repressor³¹, was downregulated in s-mNB as compared to s-iNB, consistently with the large increase in DSG at the transition between s-iNB and s-mNB.

Altogether, these data confirm at the molecular level that s-iNB comprises cycling progenitors at the transition between s-TAP and s-mNB that undergo important molecular switches, which include major changes in the expression of some RSR genes leading to a major increase in the number of DSG occurring at the final maturation of s-mNB.

s-iNB correspond to specific SVZ cell clusters identified by single cell RNA sequencing (scRNAseq)

We compared our microarray datasets to published scRNAseq datasets from adult mouse SVZ^{14,17,32}. Violin plots show that s-qNSC and s-mNB perfectly matched with the corresponding populations identified in these studies, which was less obvious for s-aNSC and s-TAP (**Fig. S5**). s-iNB matched with the clusters termed “Mitosis” (Class VI) in the study of Llorens et al.³² and “Mitotic TAP” in that of Zywitz et al.¹⁷, and with clusters 8, 10, 17, 16 linked to “Mitosis”, but also with clusters 6 and 15 described as DCX⁺Ki67⁺ neuroblasts in the study by Cebrian_Silla et al.¹⁴. These comparisons thus confirm that s-iNB comprise cycling SVZ cells exhibiting both TAP and NB features, which illustrates the difficulties to determine the exact nature of cell clusters identified by scRNAseq.

We have previously reported²⁰ that after brain exposure to ionizing radiation, proliferating progenitors are highly radiation-sensitive and rapidly eliminated after irradiation. On the contrary qNSC that are radiation resistant repopulate the different SVZ neurogenic cell populations successively the following days after irradiation. We therefore used this model and our transcriptomic signatures of SVZ stem and progenitor cells to gain new insights on the nature and the hierarchical ordering of neural progenitor clusters identified by sc-RNAseq. TenX Genomics droplet-based single-cell transcriptomic analyses were performed on SVZ cells from 2 mice, 5 days after irradiation (4 Gy, SVZ only) and from 2 non-irradiated control mice after removal of myelin and red blood cells (**Fig. 3A**). Based on a filtering criterion of 1,500 genes/cell to exclude low-quality sequenced cells, 17,343 cells were retained for analysis, divided into 11,529 non-irradiated cells and 5,814 irradiated cells (**Fig. S6**).

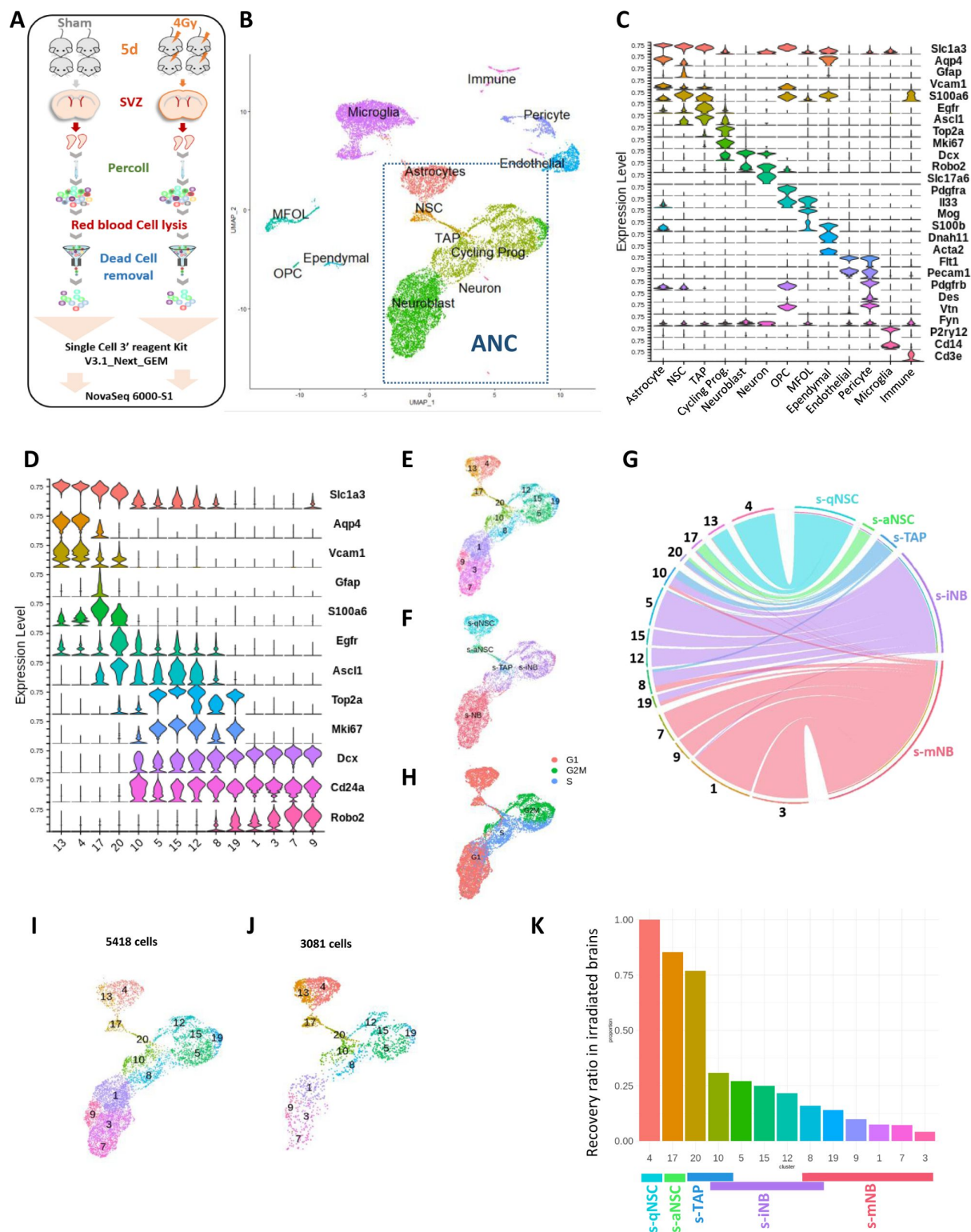


Figure 3

scRNA-seq analysis: whole SVZ

(A) Experimental design. (B) Uniform Manifold Approximation and Projection (UMAP) of the 17343 high-quality sequenced cells annotated with corresponding cell type, combining scRNA-seq datasets of non-irradiated mice and 4Gy irradiated mice. Clustering analysis at resolution 1.2 segregated cells into 33 clusters that were matched with corresponding cell types determined with marker expression in (C). Among them, 14 clusters (dotted lines rectangle) corresponded to the clusters of astrocytes and neurogenic cells (ANC). (C) Violin diagrams illustrating the expression level of known cell markers in the 33 clusters according to the literature. (D) Violin diagrams representing the expression level of selected markers specific to neurogenic cells and astrocyte from ANC clusters at the resolution 1.2. (E) UMAP focusing on the ANC subset. (F) UMAP of the top UCell score for the top 100 highly expressed genes of each neurogenic cell sorted population in the ANC clusters (Supp_data_1). (G) Chord diagram illustrating the correspondence of cell clusters with cell top score. Cluster 10 matched with both s-TAP and s-iNB and clusters 8 and 19 both to s-iNB and to s-mNB. (H) Feature plot of cell cycle scoring. (I, J) UMAP representation of ANC clusters in the unirradiated (I) and irradiated (J) samples. (K) Barplot showing the recovery ratios of each ANC cluster in irradiated brains calculated as the number of cells per clusters from irradiated brains / control brains normalized to the respective numbers of s-qNSC (cluster 4). Taking into account UCell score results, s-iNB clusters overlapped the cluster 10 and 8 containing s-TAP and s-mNB, respectively.

Non-irradiated and irradiated samples were combined and integrated using the SCTransform workflow. After integration, PCA was performed and 50 PCs were used for Uniform-manifold approximation (UMAP). Leiden graph-based clustering, with a resolution of 1.2, identified 33 clusters (Fig. S7A). These clusters were first annotated based on the expression of known cell markers (Fig. 3B and C, Fig. S7B). Nineteen clusters gathering more than half SVZ-residing cells corresponded to non-neurogenic cells (ordered by decreasing abundance): microglia (expressing *P2ry12*³³), ependymal cells (expressing both *S100β* and *Slc1a3*³⁴), endothelial cells (expressing *Flt1*³⁵), oligodendrocyte progenitors (OPC, expressing *Pdgfra*³⁶), Myelin forming oligodendrocytes (MFOL, expressing *Mog*¹⁷) and pericytes (expressing both *Vtn* and *Pcam1*^{7,37}), and dopamine neurons (expressing *Vglut2*³⁸). The 14 other clusters corresponded to astrocytes and neurogenic cells (Fig. 3B, C, D). The clusters 4 and 13 that expressed *S100β* were further identified as qNSC (cluster 4) and astrocytes (cluster 13) based on the transcriptomic signatures described by Cebrian-Silla et al., 2021¹⁴ (Fig S7C, Supp_data_5). Cluster 17, was identified as aNSC based on *Gfap*, *Thbs4*, *S100a6*, *Egfr*^{low} and *Ascl1* expressions¹⁴, and cluster 20 as TAP based on *Egfr* and *Ascl1* expressions³⁹. Clusters 10, 5, 15, 12, and 8 were defined as cycling progenitors based on the expression of proliferative markers such as *Top2a*, *Mki67*, *Ascl1*. Clusters 1, 3, 7 and 9 were identified as mNB due to the loss of *Mki67*, *Top2a* and *Ascl1* expressions and the expression of *Robo2* and *Dcx*. Cluster 19 did not expressed *Ascl1* but *Top2a* and *Mki67* as well as with *Robo2* and *Dcx*, was therefore positioned at the transition between iNB and mNB.

We then applied the UCell workflow to further characterize the neurogenic clusters with the transcriptomic signatures of each of our five sorted SVZ cell populations (Supp_data_1), calculating individual cell scores for each signature based on the Mann–Whitney U statistical analysis, and attributing cells to populations based on the highest score amongst signatures⁴⁰. This score allowed us to unequivocally attribute some clusters to the sorted cell populations (Fig. 3E, F and G): cluster 4 and 13 matched with s-qNSC, cluster 17 with s-aNSC, cluster 20 with s-TAP, clusters 5, 15 and 12 with s-iNB and finally clusters 7, 9, 1 and 3 to s-mNB. Reflecting the continuous nature of the SVZ neurogenic process, the cluster 10 matched both with s-TAP and s-iNB and clusters 8 and 19 both to s-iNB and to s-mNB. Cell-Cycle Scoring³⁶ indicated that the clusters related to s-iNB were enriched in cells in S or G/2M phase (Fig. 3H).

As the consequence of radiation-induced cell death, the relative abundances of each cluster were different in controls (**Fig. 3I**) and irradiated (**Fig. 3J**) brains and dependent on the progressive repopulation of the different types of progenitors. Normalized to the numbers of s-qNSC (cluster 4), which are supposed not to be affected by irradiation²⁰, the ratios of the number of cells per clusters in irradiated brains over that in unirradiated controls gave an index of the progressive reconstitution of each cluster (**Fig. 3K**). As expected, clusters 17 and 20, corresponding to s-aNSC and s-TAP respectively, were almost entirely regenerated. The decreasing values of the ratios evidence a temporal ordering of the regeneration of each cluster corresponding to s-iNB along with neurogenesis as follows: 10, 5, 15, 12, and 8, the latter being the one at the iNB vs mNB transition as anticipated above.

We next addressed the expression of RSR genes in each cluster (**Fig. 4**). Not all genes could have been investigated since some of them were below the detection threshold in all the cell populations studied. The expression of RSR genes (*Kdrbs3*, *Kdrbs1*, *Elavl3* and *Elavl2*, *Elavl4*, and *Celf1*) that were shown to increase along with neurogenesis in microarrays, also increased in cluster 10, remained stable in the other clusters related to s-iNB and reached a maximum value in the s-mNB clusters (**Fig. 4A**). Conversely, the expressions of *Mbnl1* and *Msi2* that were shown to decrease along with neurogenesis in microarrays (**Fig. 4B**) peaked in the clusters related to s-aNSC (17) and s-TAP (20), decreased in the cluster 10, remained low in the clusters related to s-iNB (5, 15, 12, 8) and were not detected in the clusters related to s-mNB (19, 1, 3, 7, 9). Altogether, these data corroborate our microarray data and show that a major switch in the expression of RSR genes occurs in the clusters related to s-iNB.

Interestingly, the splicing repressor *Pthp1* was similarly expressed in s-aNSC, s-TAP and s-iNB-related clusters but was downregulated progressively in the successive s-mNB clusters, suggesting that changes in *Pthp1* expression constitute an important molecular step for the final maturation of mNB, which involves a major increase of DSG.

The *Dcx* status identifies two molecularly distinct subpopulations in s-iNB

We have previously shown by immunofluorescence that not all the s-iNB were positive for doublecortin (*Dcx*) expression, a common marker of neuroblasts²⁰. Violin plot (**Fig. 3D**) and feature plot (**Fig. 5A**) show that *Dcx* expression was restrained to s-mNB and to clusters related to s-iNB in which it increased consistently with their hierarchical ordering we defined above. Variations in *Dcx* expression among iNB related clusters led us to analyse separately *Dcx*^{low} and *Dcx*^{high} cells in these clusters (**Fig. S8**). The ratios of numbers of cells per cluster indicated a greater recovery of *Dcx*^{low} cells as compared to *Dcx*^{high} cells in s-iNB related clusters from irradiated brains (**Fig. 5B**), reflecting their temporal ordering.

Characterization of differentially expressed genes according to the *Dcx* status in these clusters showed 588 unique genes expressed in the *Dcx*^{low} cells and 98 genes specifically expressed in the *Dcx*^{high} cells. Functional enrichment profiling using the g:Profiler software illustrates biological differences between both populations : genes related to Metabolism of RNA were overrepresented in the *Dcx*^{low} cells whereas genes linked to GABAergic synapse, Neuronal System were rather overexpressed in the *Dcx*^{high} cells (Supp_data_6). Notably, 55% of *Dcx*^{high} cell specific genes were s-mNB genes (e.g. *Robo2*, *Gap43*, *Nav3*). Further analysis of RSR genes expression in function of the *Dcx* status, clearly confirms that progressive transcriptomic changes occurred between *Dcx*^{low} and *Dcx*^{high} cells (**Fig. 5C** and **D**). However, RSR gene profiles of *Dcx*^{low} cells in clusters related to s-iNB were obviously different from that related to both s-TAP and s-NSC, and that of *Dcx*^{high} cells from that of s-mNB. Similar results were obtained with cells from irradiated brains (**Fig. S9**).

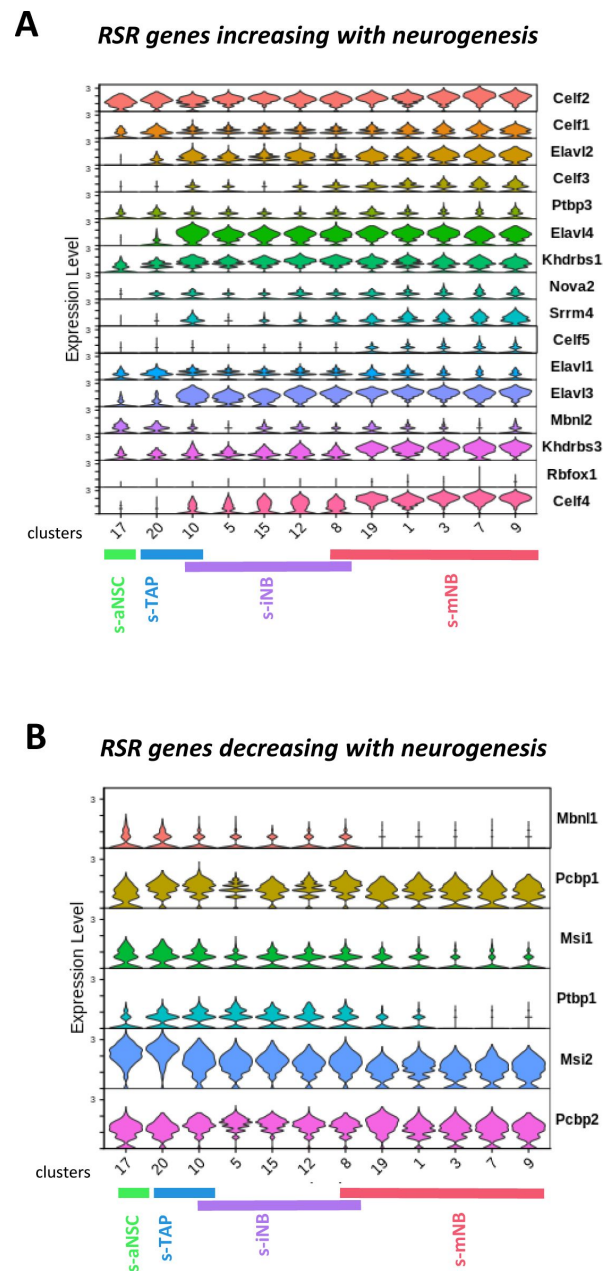


Figure 4

Expression of RSR genes in scRNA-seq neurogenic clusters

Violin diagrams illustrating the expression of the RSR genes increasing (**A**) or decreasing (**B**) with neurogenesis.

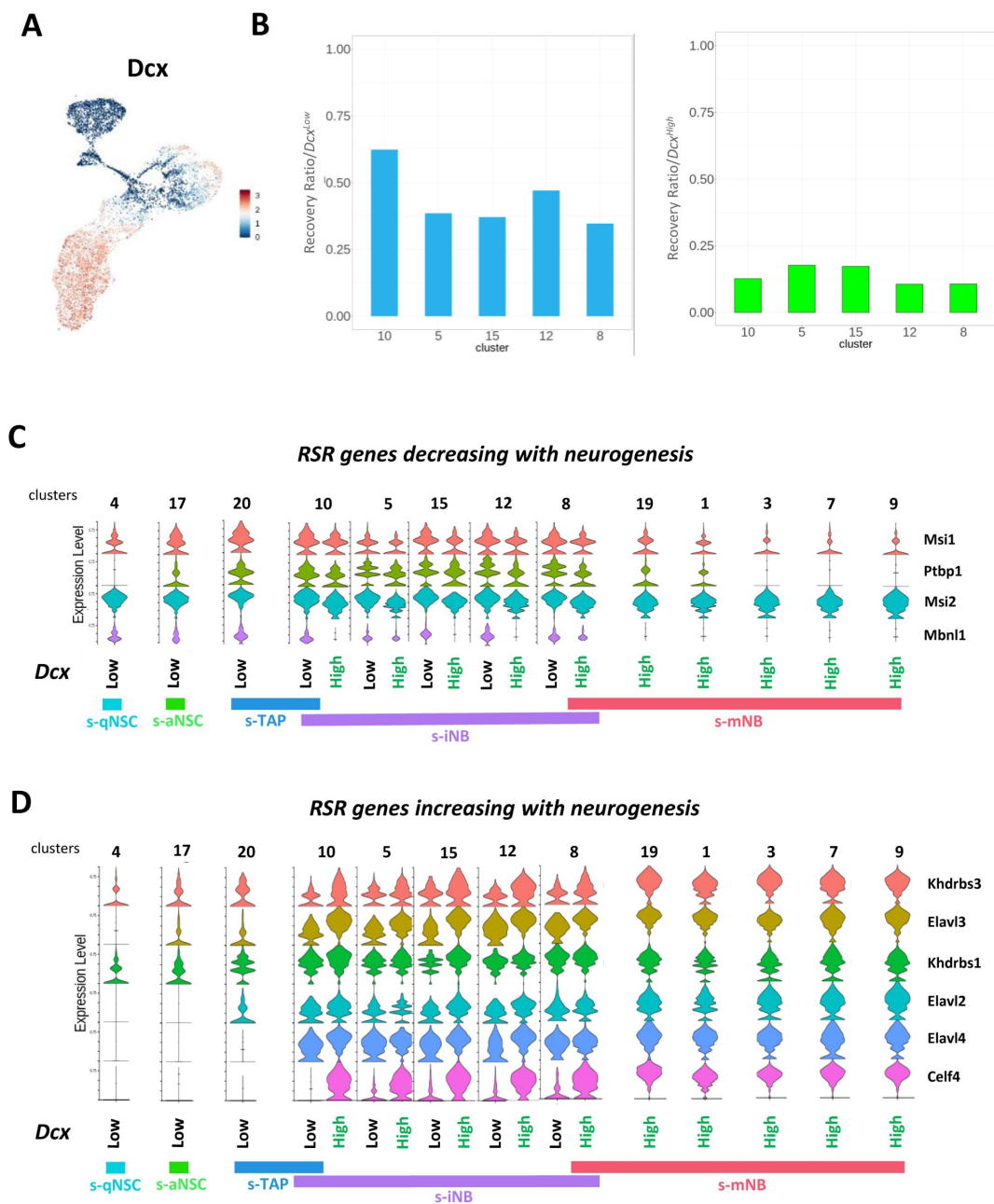


Figure 5

Dcx expression in the neurogenic cell clusters

(A) Feature plot. (B) Barplot representations of the recovery ratios of Dcx^{low} (left) and Dcx^{high} (right) in the iNB clusters of the irradiated brains, calculated as in [Figure 3K](#). Violin diagrams illustrating the expression level of RSR genes decreasing (C) and increasing (D) with neurogenesis in Dcx^{low} and Dcx^{high} in neurogenic cell clusters.

These data clearly show that the clusters related to s-iNB contained cells engaged in neuronal differentiation that are molecularly distinct from the other types of SVZ neurogenic cells already characterized. This suggests that s-iNB correspond to a new stage of cycling neurogenic progenitors progressively engaged in neuronal differentiation.

We next took advantage of the DCX-CreERT2::CAG-floxed-eGFP mice model, allowing the expression of eGFP in cells expressing *Dcx* after the administration of Tamoxifen to sort eGFP⁺CD24⁺EGFR⁺ cells corresponding to *Dcx*^{high} iNB and eGFP⁺CD24⁺EGFR⁺ cells containing the *Dcx*^{low} iNB to distinguish further these two populations at the functional level. As shown in **Fig. 6A** and **B**, recombinant eGFP⁺ iNB exhibited a clonogenic potential and a rate of population doublings quite similar to eGFP⁺ iNB and total iNB, contrary to eGFP⁺ mNB, which did not show clonogenic potential and did not proliferate at long-term *in vitro*. Moreover, we showed that both eGFP⁺ iNB and eGFP⁺ iNB have the capacity to generate the three neural lineages, neurons, astrocytes and oligodendrocytes, when plated for 5–7 day in the appropriate differentiation media (**Fig. 6C**), similarly as s-iNB (**Fig. 1C**).

By contrast, when eGFP⁺iNB were transplanted into the brain of C57Bl/6J mice, no eGFP⁺ cells persisted at long-term (5 weeks) (**Fig. S10**), suggesting that the *in vivo* plasticity and regenerative properties of s-iNB was rather associated to *Dcx*^{low} iNB.

Discussion

This study bridges the cellular and molecular characterizations of SVZ neural progenitor populations. We combined the transcriptional profiling using DNA microarrays of SVZ that were sorted by an efficient FACS strategy¹⁹ and large-scale scRNAseq to decipher the progressive molecular and cellular changes involved in SVZ neurogenesis. Our model of regeneration of the neurogenic niches after brain irradiation²⁰ allowed a unique way to determinate the sequential regeneration of the different types of neurogenic cell clusters identified by scRNAseq that were precisely characterized using the respective molecular signatures of sorted-SVZ cells, providing new insights onto the progression of SVZ neurogenesis. We show that iNB is an abundant population of cycling progenitors, which is more advanced towards neuronal differentiation than TAP, while retaining unexpected stem cell properties unlike mNB. We suggest that major splicing regulations in iNB might be critical for the final commitment to the neuronal fate.

Previous studies of the different sub-populations of SVZ progenitors were carried out using transcriptomic approaches based on the expression of various more or less specific markers. These approaches enabled the identification of quiescent and activated neural stem cells as well as mature neuroblasts, but were confronted with the strong influence of the cell cycle on cell clustering. Indeed, neural progenitors in these studies cycling have been gathered in either “mitotic” clusters^{14,17,32} or “neural progenitor cells” clusters¹⁵ that lacked clear biological significance and prevented the identification of subtypes of SVZ cycling progenitors. Our study, combining for the first time characterization of FAC-sorted cells and an irradiation-based model of sequential regeneration, clearly distinguished the molecular profiles of TAP and iNB among cycling progenitors reflecting differences in their respective potentials *in vitro* and *in vivo*.

Our results constitute thus a valuable resource for delineating the molecular transitions occurring during neurogenesis that are orchestrated by numerous differentially expressed genes and alternative splicing variants. The transition from qNSC to aNSC has been abundantly investigated in the literature^{15,32}, but the transcriptional shift leading to the differentiation of neural progenitors into neuroblasts has not been thoroughly analysed to date. We showed that it occurs in immature neuroblasts, which form a relatively abundant subpopulation of SVZ cells, which exhibit specific transcriptional features distinguishing them from TAP and mNB.

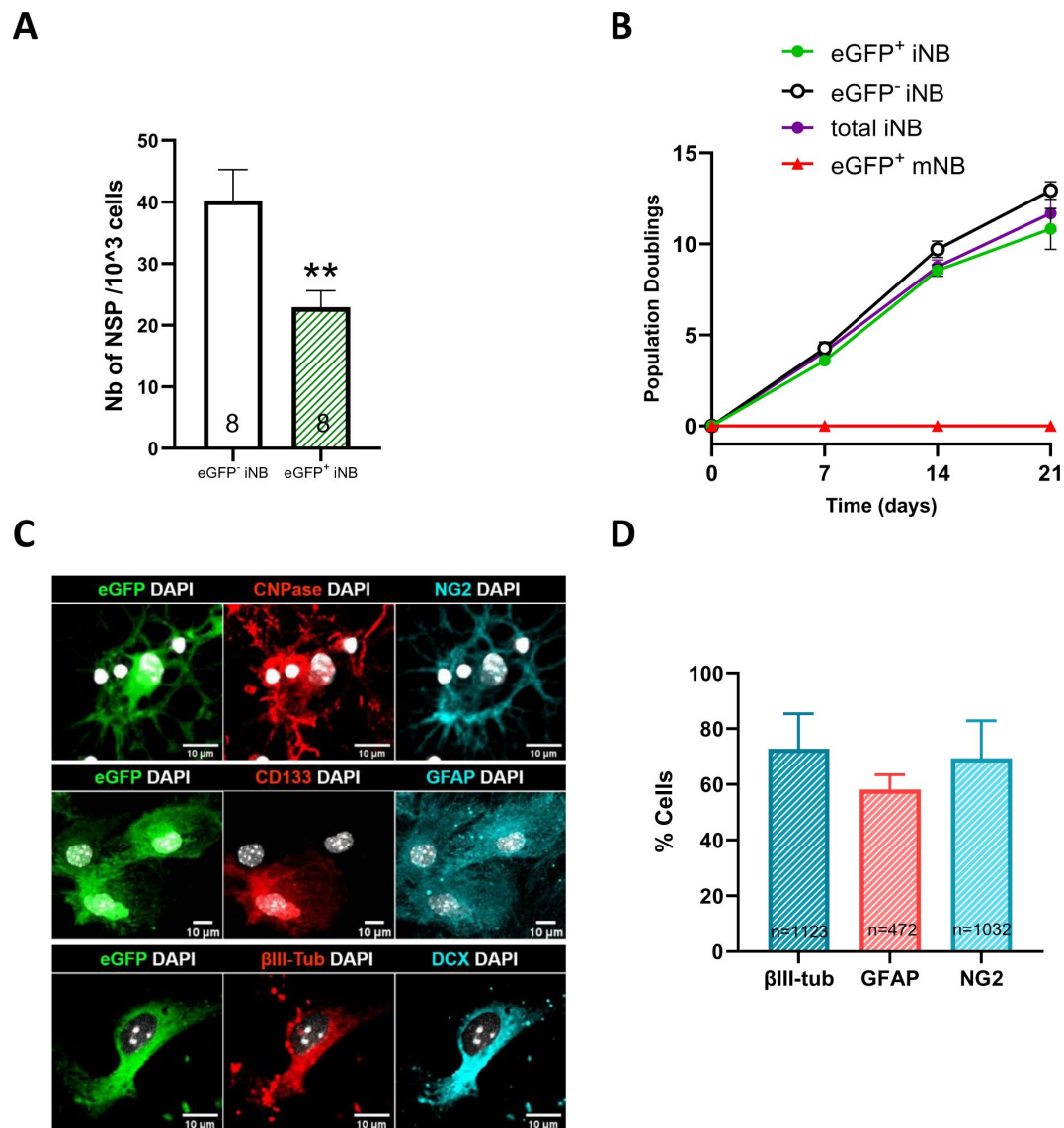


Figure 6

Plasticity of *Dcx*^{High} iNB *in vitro*

(A) Clonogenic potential of FACS-isolated eGFP⁺iNB and eGFP⁻iNB, and recombined eGFP⁺mNB isolated from DCX-CreERT2::CAG-floxed-eGFP mice, 7 days after the first injection of tamoxifen. **(B)** Population doublings (PD) of FACS-isolated eGFP⁺iNB and eGFP⁻iNB, eGFP⁺mNB cells and total iNB cells (regardless of eGFP status). No statistical difference was found in the growth rate of the different isolated populations. (n = 8) **(C)** Representative images of immunofluorescence of freshly sorted eGFP⁺iNB cells cultured in oligodendrocyte, astrocyte, or neuronal differentiation medium, and stained for NG2 and CNPase, GFAP and CD133, βIII-Tubulin and DCX, respectively. Scale: 10 μm. **(D)** Quantification of the percentage of eGFP⁺ iNB among DAPI cells cultured in differentiation media that differentiate in either neurons (βIII-tubulin), oligodendrocytes (NG2) or astrocytes (GFAP).

We formerly described the isolation of s-iNB as a subset of cycling SVZ cells that expressed both CD24, a neuroblast marker, and EGFR, a marker of cycling progenitors^{19,21}. This was consistent with prior findings describing a subset of neuroblasts co-expressing *Ascl1* and *Dcx* or CD24^{low} expressing low levels of EGFR in the adult SVZ^{13,39,43}. Here, we evidenced that s-iNB comprise a molecularly distinct population of cycling cells at the transition between TAP and mNB, and exhibiting molecular characteristics of both these cell types.

The capacity of neuroblasts to reorient towards the glial lineage in pathological contexts have been reported previously^{44–46}. Consistent with the molecular findings showing that s-iNB kept transcriptional features of NSC and TAP, we showed that s-iNB also kept *in vitro* and *in vivo* a regenerative potential similar to that of s-aNSC and s-TAP. Importantly, this may also include the capacity to dedifferentiate *in vivo* and generate NSC as suggested by the detection of GFAP-positive engrafted cells persisting in the SVZ.

Interestingly, s-iNB corresponded to several scRNAseq clusters, which were not clearly characterized in the literature. Our model of SVZ regeneration established the hierarchical ordering of these clusters, in which we showed progressive molecular changes leading to the differentiation into migrating neuroblasts. This includes the increase in *Dcx* expression. *Dcx*^{high} cells in these clusters have transcriptional features confirming they are more engaged in the differentiation than their *Dcx*^{low} counterparts. Importantly, we showed that *Dcx*^{high} and *Dcx*^{low} cells in these clusters remained molecularly distinct from both TAP and mNB. We thus propose a revision of the current model of SVZ neurogenesis by introducing iNB cells, as highly cycling progenitors that undergo progressive neuronal differentiation but kept multipotency (**Fig. 7**). iNB cells could be further divided in two subclasses of cells depending on their progression into the neuronal differentiation process: iNB1 cells, corresponding to *Dcx*^{low} iNB, immediately succeeding the TAP and iNB2 cells, corresponding to *Dcx*^{high} iNB and preceding the stage mNB. Importantly, we showed that both iNB1 and iNB2 cells kept transcriptomic features that differentiated them from both TAP and mNB, whereas iNB1 cells kept a regenerative potential both *in vitro* and *in vivo*.

One of the most important findings of this study is the first characterization of AS events and RSR expression variations occurring along with neurogenesis. AS events are known to play important roles in neural cell differentiation, neuronal migration, synapse formation, and brain development^{27,30,47–49}. Neurons are characterized by a pervasive use of alternative exons^{50,51}. Consistently, we showed a dramatic increase in DSG at the transition between s-iNB and s-mNB. These alternative splicing events are clearly related to major changes in RNA splicing regulators (RSR) expressions occurring in s-iNB. Indeed, we identified RSR genes whose expression increased with neurogenesis and others whose expression decreased with neurogenesis. Consistently, some genes upregulated in s-aNSC, s-TAP, such as *Msi1* and *Msi2*, have a known role in maintenance and renewal of stem cells^{52–54}, while some others upregulated in s-mNB, such as *Elavl3* and *4*, have known roles in many neuronal processes^{54,55}. Interestingly, genes of both groups of genes were overexpressed in scRNAseq clusters related to s-iNB, which thus undoubtedly distinguished these cells from both s-TAP and s-mNB and confirmed that important molecular switches for neuronal differentiation occurred in D cells.

Further analysis of scRNAseq clusters showed that these transcriptomic switches occurred sequentially from iNB1 cells to iNB2 cells. Indeed, the expression of *Elavl4* progressively increased from the *Dcx*^{low} iNB1 cells to the *Dcx*^{high} iNB2 cells fractions, confirming their progressive commitment into neuroblasts, and overall, the interest of discriminating these two cell types in the model of SVZ neurogenesis we propose (**Fig. 7**). Strikingly, the splicing repressor, *Ptbp1*, was expressed in iNB1 cells at levels similar to those in s-aNSC and s-TAP, but decreased in iNB2 cells, and was undetectable in migrating neuroblasts. This suggests that expression of *Ptbp1* may play an important role in maintaining the regenerative properties in iNB1 cells, which should be further investigated.

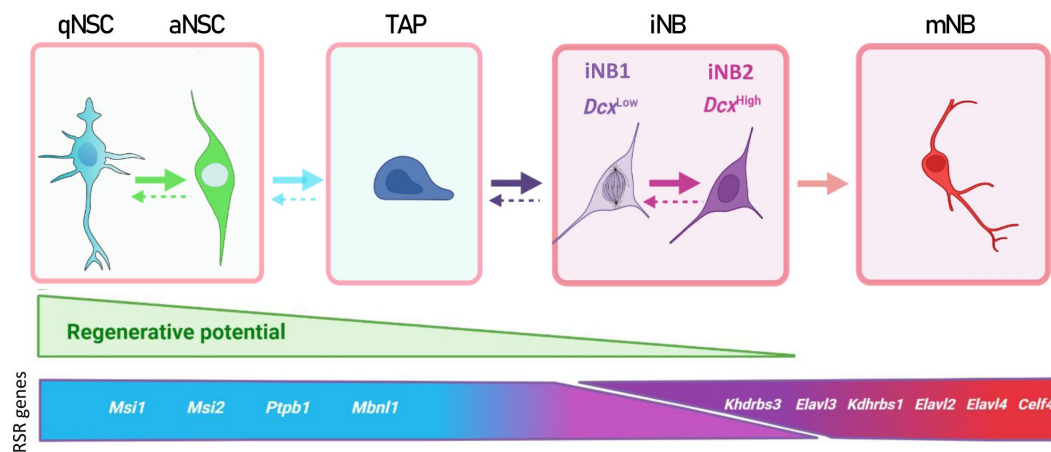


Figure 7

Proposed model of adult mouse SVZ neurogenesis involving major changes in RSR genes expression

Possible interconversion of iNB states would require further experimental confirmation.

Our study has been performed on cells collected from the SVZ the lateral walls of the ventricles of young adult mice. The transcriptional signatures we provide here may thus be useful for further characterization of neurogenic cells present in the septal wall of the SVZ¹⁸.

To conclude, we have characterized the molecular features of iNB cells, a relatively abundant population in the adult SVZ that kept unexpected plasticity and multipotency that persist in the aged brain²¹. They may thus represent a new target in regenerative medicine. We have shown that these cells undergo major transcriptomic changes involving AS and associated to switches in RSR genes expression. Dysregulations of many of these RSR genes have been involved in cancer and neurodegenerative diseases⁵⁶. Further characterization of iNB cells using the tools we developed may be useful to determine the potential therapeutic targeting of these cells in various brain pathologies.

Acknowledgements

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Author contributions

Corentin Bernou: Collection and assembly of data, data analysis and interpretation, manuscript writing.

Marc-André Mouthon: Collection and/or assembly of data.

Mathieu Daynac: Collection and/or assembly of data.

Thierry Kortulewski: Collection and/or assembly of data.

Benjamin Demaille: Analysis of Clariom datasets.

Vilma Barroca: Intracerebral neural progenitor grafting in mouse models.

Sébastien Couillard-Despres: for DCX-CreERT2:CAG-CAT-eGFP mice.

Nathalie Déchamps: Cell sorting by flow cytometry.

Véronique Ménard: Irradiation of animals.

Léa Bellenger: Analysis of scRNAseq datasets.

Christophe Antoniewski: Analysis of scRNAseq datasets.

Alexandra Chicheportiche: Conception and design, collection and assembly of data, data analysis and interpretation, manuscript writing.

François D. Boussin: Supervision of the study, conception and design, financial support, data analysis and interpretation, manuscript writing.

STAR Methods

Experimental model and subject details

Mice

All experimental procedures complied with European Directive 2010/63/EU and the French regulations. The protocols were approved by our institutional ethical committee (CEtEA 44) and authorized by the « Research, Innovation and Education Ministry » under registration number APAFIS#25978-2020061110358856 v1. Adult male C57BL/6J were obtained from Janvier Laboratories. Adult male β -actin:eGFP (Okabe et al., 1997) and DCX-CreERT2::CAG-CAT-eGFP mice kindly given by Pr Couillard-Despres Sebastien ⁵⁷[\[link\]](#). Experiments were performed with male animals aged 8 and 10 weeks. Tamoxifen (TAM, T-5648, Sigma-Aldrich) was dissolved in corn oil (C-8267, Sigma-Aldrich) at 40 mg/ml. To analyse the expression of the *Dcx* reporter gene, 20 mg TAM/g body weight was administered twice daily by gavage over a period of five consecutive days.

Cell sorting

Lateral ventricle walls of the SVZ were microdissected and dissociated as previously described ¹⁹[\[link\]](#)–²¹[\[link\]](#) with the modification that cells were centrifuged at 250g for 20 min at 4°C without brake on a 22% Percoll gradient (GE Healthcare) to remove myelin prior to single cell suspension labelling. The antibodies to distinct cell populations were anti-CD24 phycoerythrin [PE]-conjugated (Rat IgG2b; BD Biosciences, 1:50, RRID:AB_2034001), CD15/LeX fluorescein isothiocyanate-[FITC] conjugated (clone MMA, mouse IgM; BD Biosciences, 1:50, RRID:AB_400103), CD15/LeX-Brilliant violet-421-[BV421] (Mouse IgG1, κ ; BioLegend, 1:50, RRID:AB_2566519) and Alexa647-conjugated epidermal growth factor (EGF) ligand (Clone 30H45L48, Thermo Fisher Scientific, 1:200, RRID:AB_2662334). Cells were gated following the fluorescence minus one (FMO) control. Immediately prior to FACs sorting, propidium iodide (PI) or Hoechst 33258 was added to a final concentration of 1 μ g/ml to label the dead cells. Cells were sorted on an INFLUX cell sorter equipped with an 86 μ m nozzle at 40 psi (BD Biosciences). All the data were analysed using FlowJo software (Tree Star, Ashland, OR).

Cell culture expansion and differentiation

Clonogenicity assay and population doublings: FACs-purified populations were collected in NeuroCult medium complemented with the proliferation supplement (STEMCELL Technologies), 2 μ g/mL heparin, 20 ng/mL EGF and 10 ng/mL FGF-2, at a density of 1×10^3 cells/well in 96-well tissue culture plates coated with poly-D-lysine (Merck Millipore). Five days after plating, the neurospheres were counted to determine the clonogenicity of each cell population.

Neurospheres were then mechanically dissociated and sub-cultured in 24-well plates over 3 weeks to measure the population doublings at each passage (1 passage/week).

Differentiation assays: just after sorting, cells were placed on Poly-L-ornithine (P4957, Merck Millipore) coated 24-well culture plates at 40,000 cells/ml in DMEM:F12 with several supplementation according to the lineage: astroglial : 2% B27 MAO (minus antioxidant, Thermo Fisher Scientific) and 2% fetal bovine serum (10082139, Thermo Fisher Scientific), oligodendroglial : 10 ng/mL FGF-2, a defined hormone mix including Glucose (G-7021, Sigma-Aldrich 50g/ml) / NaHCO₃ (S-5761, Sigma-Aldrich, 10mg/l) / HEPES (H-3784, Sigma-Aldrich, 1.3mM), insulin (I-1882, Sigma-Aldrich, 22.5mg/l), Apo transferrin (T-2252, Sigma-Aldrich, 0.09g/l), progesterone (P-6149,

Key resources table

REAGENT or RESSOURCE	SOURCE	IDENTIFIER
Antibodies		
PE-rat Anti-mouse CD24 (clone 1/69)	BD Biosciences	Cat#307567;RRID:AB_2034001
FITC-rat Anti-mouse CD15 (clone MMA)	BD Biosciences	Cat#340703; RRID:AB_400103
BV421-mouse Anti-human CD15 (Clone W6D3)	BioLegend	Cat#323039; RRID:AB_2566519
Alexa647-conjugated epidermal growth factor (EGF) ligand (clone 30H45L48)	ThermoFisher Scientific	Cat#MA7-00308-A647
Goat Anti-GFP	Abcam	Cat#ab6673
Rabbit Anti-GFP	Abcam	Cat#ab290
Mouse Anti-mouse O4 (clone 81)	Merck Millipore	Cat#MAB345
Rabbit Anti-mouse NG2	Merck Millipore	Cat#AB5320
Goat Anti-mouse Dcx C-18	Santa Cruz	Cat#sc-8066
Rabbit Anti-mouse Dcx	CellSignaling	Cat#4604
Rabbit Anti-mouse Tubulin β 3 (TUBB3)	Covance	Cat#PRB-435P
Mouse Anti-mouse Gfap (clone GA5)	Merck Millipore	Cat#MAB3402
Rabbit Anti-mouse Gfap	Merck Millipore	Cat#G9269
Rat Anti-CD133 (clone 13A4)	Merck Millipore	Cat#MAB4310
Mouse Anti-mouse CNPase (clone 11-5B)	Merck Millipore	Cat#MAB326
Goat Anti-mouse Olig2	Biotechne	Cat#AF2418
Mouse Anti-mouse NeuN (clone A60)	Merck Millipore	Cat#MAB377
Alexa Fluor 594_ Donkey anti-Mouse IgG (H+L) Secondary Antibody	ThermoFisher Scientific	Cat#A21203
Alexa Fluor 488_ Donkey anti-Mouse IgG (H+L) Secondary Antibody	ThermoFisher Scientific	Cat#A21202
Alexa Fluor 647_ Donkey anti-Mouse IgG (H+L) Secondary Antibody	ThermoFisher Scientific	Cat#A32787
Chemical, peptides, and recombinant proteins		
Albumin Bovine Serum 10%	Merck Millipore	Cat#9048-46-8
Percoll	Merck Millipore	Cat#GE17-0891-01
DPBS w/o calcium w/o magnesium	ThermoFisher Scientific	Cat#12559069
NeuroCult Basal medium	STEMCELL Technologies	Cat#05700
Heparin Solution	STEMCELL Technologies	Cat#07980
Epidermal Growth Factor Protein, human recombinant	Merck Millipore	Cat#GF144
Fibroblast growth Factor basic, human recombinant	Merck Millipore	Cat#GF003AF
Poly-L-ornithine solution	Merck Millipore	P4957
Poly-D-lysine solution	Merck Millipore	25988-63-0
DMEM:F12 medium	ThermoFisher Scientific	31331028
B-27-Supplement Minus Antioxydants (AO)	ThermoFisher Scientific	Cat#10889038
Fetal Bovine serum	ThermoFisher Scientific	Cat#10082139
D-(+)-Glucose	Merck Millipore	Cat#G7021
Sodium bicarbonate	Merck Millipore	Cat#S5761
HEPES sodium salt	Merck Millipore	Cat#H3784
Insulin from bovine pancreas	Merck Millipore	Cat#I1882
Apo-Transferrin human	Merck Millipore	Cat#T2252
Progesterone	Merck Millipore	Cat#P6149
Putrescine	Merck Millipore	Cat#P7505
Sodium selenite	Merck Millipore	Cat#S9133
Papain	Worthington Biochemical	Cat#LK003150

Tamoxifen	Sigma Aldrich	Cat#T5648
Corn oil	Sigma Aldrich	Cat#C8267
Solution de paraformaldéhyde, 4 % en PBS	ThermoFisher Scientific	Cat#15670799
Triton X-100	Sigma Aldrich	Cat# 11332481001
Propidium iodide	Sigma Aldrich	Cat#P4170
Hoechst 33258	ThermoFisher Scientific	Cat#H3569
Critical Commercial Assays		
Red blood Cell lysis solution	Milteny Biotec	Cat#130-094-183
Dead Cell Removal Kit	Milteny Biotec	Cat#130-090-101
RNeasy Micro Kit	Qiagen	Cat#74004
RNase-Free DNase Set	Qiagen	Cat#79254
AgilentHigh Sensitivity DNA Kit	LabChip	Cat#5067-4626
Chromium Next GEM Chip G Single Cell Kit	10X GENOMICS	Cat#PN-1000127
Dual Index Kit Plate TT Set A	10X GENOMICS	Cat#PN-1000215
Deposited data		
List of Significantly Upregulated Genes in Subclusters, Related to Figures 3 and S4.	Zywitsa et al., 2018	https://doi.org/10.1016/j.celrep.2018.11.003 TableS3 File:1-s2.0-S2211124718317327-mm4.xlsx
B cells vs. astrocytes: DE genes and GO terms	Cebrian et al. 2021	https://doi.org/10.7554/eLife.67436 Supplementary file 1
scRNAseq datasets from the University of California Santa Cruz Cell Browser	Cebrian et al. 2021	https://svzneurogeniclineage.cells.ucsc.edu
List of cluster gene sets and GO terms	Llorens et al. 2015	https://doi.org/10.1016/j.stem.2015.07.002 File:1-s2.0-S193459091500301X-mm3.xls
Softwares		
RStudio	RStudio	https://www.rstudio.com/
Transcriptome Analysis Console (TAC 4.0)	ThermoFisher Scientific	-Ritchie ME et al. (2015) limma powers differential expression analyses for RNAsequencing and microarray studies. Nucleic Acids Research 43(7):e47. -Romero JP et al. (2016) EventPointer: an effective identification of alternative splicing events using junction arrays. BMC Genomics 17:467.
Other		
70µm cell strainer	ThermoFisher Scientific	Cat#10788201
2.5µl, Model 62 RN SYR, Small Removable NDL, 22s ga, 2 in, point style 3	Hamilton	Cat#87942
Small Hub 33G	Hamilton	Cat#074750

Key resources table (continued)

Sigma-Aldrich, 18 μ M), putrescine (P-7505, Sigma-Aldrich, 8.7mg/l), and sodium selenite (S-9133, Sigma-Aldrich, 4.7 μ M), and neuronal : 2% B27 MAO. Cultures in hypoxic conditions (4%O₂) at 37°C for 5 (oligodendroglial or astrocytic differentiations) or 7 days (neuronal differentiation).

Transplants

Three injections of 1 μ l of freshly eGFP⁺ cells from β -actin:eGFP or DCX-CreERT2:CAG-CAT-eGFP mice were performed in the striatum at proximity of SVZ of wild-type C57Bl6/J recipient mice using the following coordinates: (1) AP, 0.0; L, 1.4; V, -2.1; (2) AP, 0.5; L, 1.1; V, -2.2; (3) AP, 1.0; L, 1.0; V, -2.5mm relative to bregma, as described before by Codega et al. 2014²³. For CD1 mouse transplants with freshly eGFP⁺ cells from DCX-CreERT2:CAG-CAT-eGFP mice the injection coordinates were redefined from the ratio of the distance bregma to lambda for the C57Bl6/J model over the distance bregma to lambda for the CD1 model.

The transplantations were performed using a small animal stereotaxic apparatus (Kopf model 900) with a 2.5 μ l Hamilton syringe (Hamilton, Bonaduz, Switzerland). Recipient mice were sacrificed 5 weeks after transplantation.

Tissue processing, immunolabeling and microscopy

Deeply anaesthetized animals received a transcardial perfusion of 4% paraformaldehyde. Brains were post-fixed overnight in 4% PFA and cryoprotected in 30% sucrose/PBS. Serial coronal cryostat sections were made at 14 μ m of thickness (Leica CM3050S). For each brain, all the sections from the hippocampus to olfactory bulb were deposited on a set of 14 slides with a 150 μ m step. Sections underwent permeabilisation in phosphate buffered saline (PBS) containing 0.3% Triton X100 and 1% of Bovine Serum Albumin 1h at room temperature. Sections were incubated with primary antibodies PBS containing 0.1% Triton X100 overnight at 4°C. Secondary antibodies used were Alexa 594, 488 and 647 and were applied at 1:400 (Thermo Fisher Scientific) for 2h at room temperature. Each staining was replicated in at least 3 slides from different mice.

Cells were fixed with 4% PFA for 15 minutes at room temperature and rinsed three times with PBS. Staining was performed as above except that antibodies were applied for 2h at room temperature.

The following antibodies used were: rabbit anti-eGFP (Abcam, ab290, 1:300), goat anti-GFP (Abcam, ab6673, 1:300), mouse anti-O4 (Millipore, MAB345, 1:100), rabbit anti-NG2 (Millipore, AB5320, 1:100), goat anti-Olig2 (R&D, AF2418, 1:200), mouse anti-NeuN (Millipore, MAB377, 1:100), rabbit anti-Dcx (Cell Signaling, 4604, 1:400), goat anti-Dcx (Santa Cruz, sc-8066, 1:100), mouse anti-GFAP (Millipore, MAB3402, 1:400), rabbit anti-Gfap (Sigma-Aldrich, G9269, 1:400), mouse anti-CNPase (Millipore, MAB326, 1:100), rabbit anti- β -III tubulin (Covance, PRB-435P, 1:200), rat anti-CD133 (Millipore, MAB4310, 1:100).

Brightfield and fluorescent images were captured through Plan Apo X20/ Numerical Aperture (NA):1.3 oil objective and Plan Apo X40/ NA:0.75 dry objective using hybrid detection technology on a laser scanning confocal (Leica Microsystems SP8), Nikon A1R confocal laser scanning microscope system attached to an inverted ECLIPSE Ti (Nikon Corp., Tokyo, Japan)

SVZ irradiation

Animals were anesthetized with isoflurane (3% induction for 5 minutes) and 1.5-2% when the mice were placed in Small Animal Radiation Research Platform (SARRP, XSTRAHL, LTD Company). First, a Cone Beam Computer Scanning (CBCT) was performed to target the SVZ and to set up the treatment planning system. Mice were placed on their stomach and two fields (-45° and + 45°) were used to be homogeneous on the irradiation target. The size of the irradiation field was

adapted with a multivariable collimator 10*6 mm. The SVZ received a 4Gy dose distributed 50% by each beam and the dose output was around 3.64 Gy/min depending of the size of the field. The X-ray configuration was 220 kV, 13 mA and 0.15 mm of Copper in these conditions.

Single cell library preparation for single cell RNA sequencing

SVZ lateral ventricle walls were dissected from two non-irradiated mice brains (2 month old) and two 5 days post-irradiation 4Gy-irradiated mice brains (2-month old).

Protocol of SVZ digestion with Papain (Worthington, LK003150) to obtain single cell suspensions were strictly identical to those performed in the whole genome microarray experiment described above, including the Percoll gradient step to remove myelin. Cells were resuspended in 0.04% BSA in PBS passed through a 70µm cell strainer (ThermoFisher Scientific).

Blood cells were lysed using the Red blood Cell lysis solution (Milteny Biotec, 130-094-183) and dead cell Removal using the Dead Cell Removal Kit (Milteny Biotec, 130-090-101) were performed following the manufacturer's instructions.

Single cell suspensions of two replicates of each sample (unirradiated, 4Gy) were adjusted to 2000 cells/µl and 10,000 non-irradiated cells and 6,000 irradiated cells were loaded per channel onto Chromium Next GEM Chip Single Cell kit (ref: 1000127). Library preparation was performed using the Dual Index Kit TT (ref: 1000215) according to manufacturer's recommendations (10x Genomics, Pleasanton, CA). Quality and quantification of libraries were done using the High Sensitivity DNA LabChip kit (ref: 5067-4626). Sequencing NGS Standard Illumina was carried out with the NovaSeq6000-S1 flow cell (ICM platform, Paris).

Method details

Whole-genome Microarrays

mRNA were isolated with the RNeasy Micro Kit with DNase treatment (Qiagen) and sample quality was controlled using the Agilent Bioanalyser. MTA-1.0 arrays (Clariom D Affymetrix technology) were performed according to the manufacturer's protocol (ThermoFisher). PCA and pseudotime ordering was performed using TSCAN online user interface (Ji and Ji, 2016). The data were analysed with the freeware software Transcriptome Analysis Console (TAC) applying the RMA algorithm normalization. Functional profiling were generated using REACTOME and KEGG datasets obtained from the g:Profiler interface (<https://biit.cs.ut.ee/gprofiler/gost> .

Signatures and unique mRNA splicing isoforms of cell sorted populations were delineated using Venn diagrams comparing the five populations of neural progenitors (s-qNSC, s-aNSC, s-TAP, s-iNB and s-mNB) (https://bioinformatics.psb.ugent.be/cgi-bin/liste/Venn/calculate_venn.html .

sc-RNAseq and microarray dataset comparisons

Published genelists were matched with microarray data raw data using aggregated Z-scores. For each gene in a genelist, gene expression amongst all samples was normalized between 0 and 1, then mean expression among samples was calculated for each population. Plots represent the distribution of mean normalized gene expression for all genes in the genelist.

Single-cell RNA control quality and analysis

ScRNA-seq data were aligned to the GRCm38 - mm10 reference genome (cellranger-6.1.2). Data were analysed in R (version 4.1.0) using Seurat (v4.1.1). Quality control removed outlier cells with fewer than 1200 to 1500 genes per cell (nFeatures_RNA), 2350 to 2700 UMI (nCount_RNA) and cells less than 10% mitochondrial content. Gene expression was normalized using the regularized

negative binomial regression to normalize UMI count data (SCTransform) using 4000 variable features and the different samples (controls and irradiated). The features to use in the downstream integration procedure were determined using `SelectIntegrationFeatures`. After a pre-computed `AnchorSet` (`FindIntergrationAnchors`), the Seurat object were integrated using `IntegrateData` with the normalization method SCT. `CellCycleScoring` was applied to calculate the S.Score, G2M.Score and Phase. `RunUMAP` runs the Uniform Manifold Approximation and Projection (UMAP) dimensional reduction technique using 50 PCs. `FindNeighbors` computes the k.param nearest neighbors using `dims = 1:50`. Clusters of cells were identified by a shared nearest neighbor (SNN) modularity optimization bases clustering algorithm using `FindClusters` with resolutions from 0.2 to 2. The resolution 1.2 that segregates cells into 33 clusters was chosen based on Clustree graph showing the relationship between clusterings at the different resolutions. UMAP plots were generated with `UMAPPlot` with default parameters.

The `PrepSCTFindMarkers` function was used to normalize gene expression for differential gene expression analysis among clusters, which was performed using `FindAllMarkers` on the SCT assay, with significance determined using a Wilcoxon rank sum test ($p_{\text{val_adj}} < 0.05$) and $\log(\text{Fold-Change})$ threshold of 0.1. A histogram representation of *Dcx* (ENSMUSG00000031285) expression in the clusters corresponding to astrocytes and neural progenitors allowed us to place a threshold expression at 1. Cells with an expression lower than or equal to 1 correspond to *Dcx*^{Low} cells and cells with an expression strictly higher than 1 to *Dcx*^{High} cells. Then, we reapplied `PrepSCTFindMarkers` and `FindAllMarkers` to find differentially expressed genes between *Dcx*^{Low} and *Dcx*^{High} iNB.

Gene expression score

We used `AddModuleScore_Ucell` (Package *UCell* version 1.3.1) to score cells based on the expression of predefined specific gene lists of sorted s-qNSC, s-aNSC, s-TAP, s-iNB and s-mNB determined by microarrays using the Mann-Whitney U statistic⁴⁰. Each cell was then assigned an identity corresponding to the highest score amongst the signatures tested. Repartition of classes among clusters was visualized using the `Circlize` package⁵⁸.

For astrocyte and B-cell classification, we used the lists of Astrocyte and B-cell markers provided by Cebrian-Silla et al. (2021)¹⁴ to score cells from the clusters 4 and 13 between 0 (min) and 1 (max). Cells were assigned to the highest score amongst both signatures.

Gene enrichment analysis

Comparisons of the specific gene lists of *Dcx*^{High} s-iNB and *Dcx*^{Low} s-iNB were performed with Biological pathways using the `g:Profiler` interface (<https://biit.cs.ut.ee/gprofiler/gost>).

Quantification and statistical analysis

Non-parametric Mann-Whitney test was conducted using Prism 8.1.2 (GraphPad Software Inc., La Jolla, CA, RRID:SCR_002798). The statistical significance was set at $p < 0.05$. The data are expressed as the mean $\pm$ SEM.

Data availability

The datasets generated for this article were deposited online on the ANNOTARE database (E-MTAB-12265, E-MTAB-12495). Scripts used for analysis are available on <https://github.com/ACHicheportiche/Bernou2023>.

Supplemental informations

Supp_data_1: Transcriptomic signatures of s-qNSC, s-aNSC, s-TAP, s-iNB and s-mNB (Clariom assay) and corresponding `g:Profiler` Gene Ontology analyses.

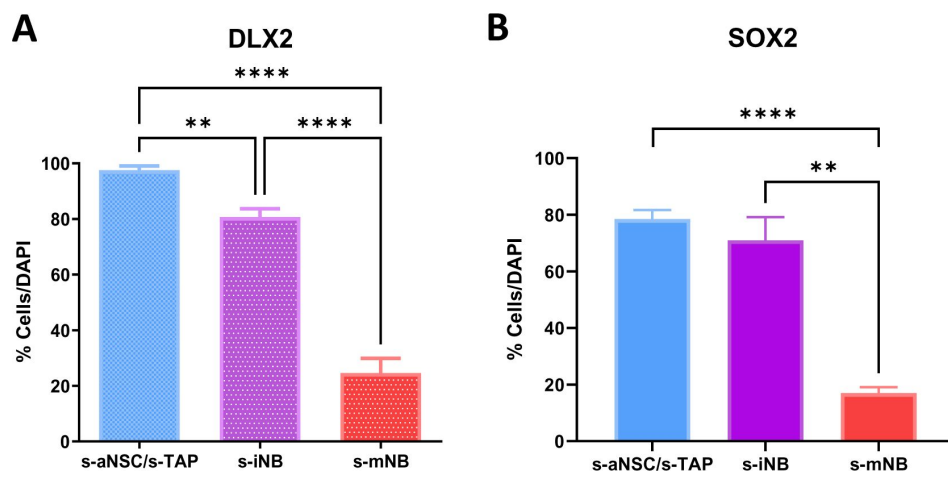
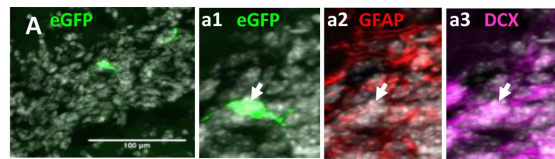


Figure S1

Expression of neural progenitor markers by SVZ cell-sorted populations

s-aNSC, s-TAP, s-iNB, and s-mNB cells were sorted from the SVZ of 2-month old C57Bl/6 mice, fixed and stained with antibodies against DLX2, a marker of proliferating progenitors (Doetsch et al. 2002) and SOX2 that plays an essential role in the maintenance of neural progenitors (Zhang et al. 2014). Quantifications of sorted SVZ cells expressing DLX2 (**A**) and SOX2 (**B**). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

A EGFP+ s-iNB cells in the rostral migratory stream



B Regenerative potential of EGFP+ s-iNB *in vivo*

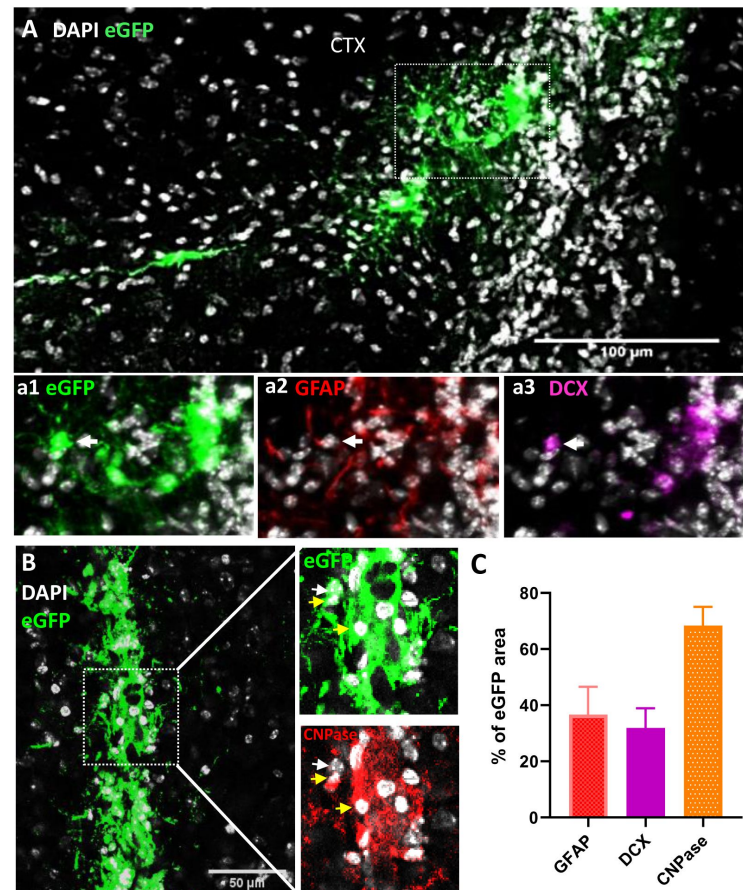


Figure S2

eGFP⁺ s-iNB cells in the rostral migratory stream and regenerative potential of eGFP⁺ s-iNB *in vivo*

Immunofluorescence of mouse brains transplanted unilaterally in 3 injection points near the dSVZ/RMS with eGFP⁺iNB cells isolated from β -actin:eGFP mice. Transplanted brains were analysed five weeks later. **(A)** Example of EGFP⁺iNB cell (white arrow) that has reached the rostral migratory stream and expressed DCX (inset a3), but not GFAP (inset a2). Insets are high magnification panels. **(B)** eGFP⁺ cells persisting at IS expressing astrocyte or oligodendrocyte markers. High magnifications of the inset1 (dotted line in A) showing eGFP⁺ GFAP⁺ cells (white arrow) (a2) and eGFP⁺DCX⁺ cells (white arrow) (a3). High magnifications of the inset2 (dotted line in B) showing eGFP⁺ CNPase⁺ (yellow arrow) and eGFP⁺CNPase⁻ (white arrow). Scale bars= 100 μ m. **(C)** Quantitative analysis of GFAP, CNPase or DCX antibodies staining of GFP-positive cells persisting at IS (where high number of grafted cells were found) in [Figure S2](#). This was performed by using the NIS software measuring eGFP-, GFAP-, CNPase- and DCX-positive areas. The intersections between each marker and eGFP areas were determined as a percentage of staining. The results showed that approximately one third of eGFP⁺ cells expressed GFAP or DCX. The quantitative analysis of CNPase expression specific to eGFP⁺ cells was complicated by host cells, but an overexpression in eGFP-positive areas clearly revealed the expression of CNPase by a significant fraction of these cells.

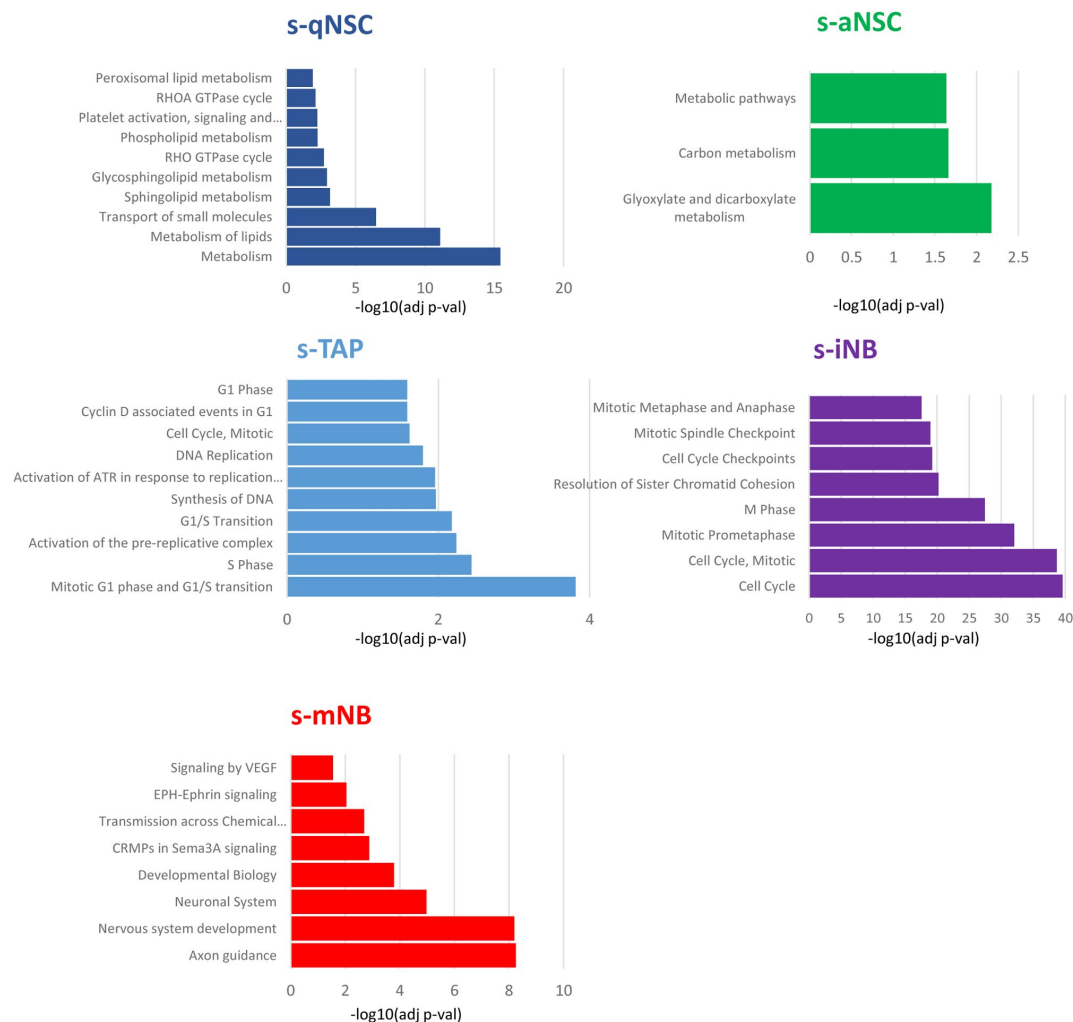


Figure S3

Functional enrichment analyses of the individual transcriptome signature of neurogenic populations

Comparisons of individual transcriptome signatures with various Gene Ontology gene sets and Biological pathways using the g:Profiler software highlight significant stage-dependant changes. S-qNSCs expressed genes related to Metabolism of lipids, Transport of small molecules. S-aNSC expressed genes involved in Glyoxylate and decarboxylate metabolism. S-TAP significantly displayed genes involved in Mitotic G1 phase and G1/S transition and S Phase. S-iNB specifically expressed Cell Cycle and Mitotic genes in contrast to s-mNB which are characterised by the expression of Axon guidance and Nervous system development genes.

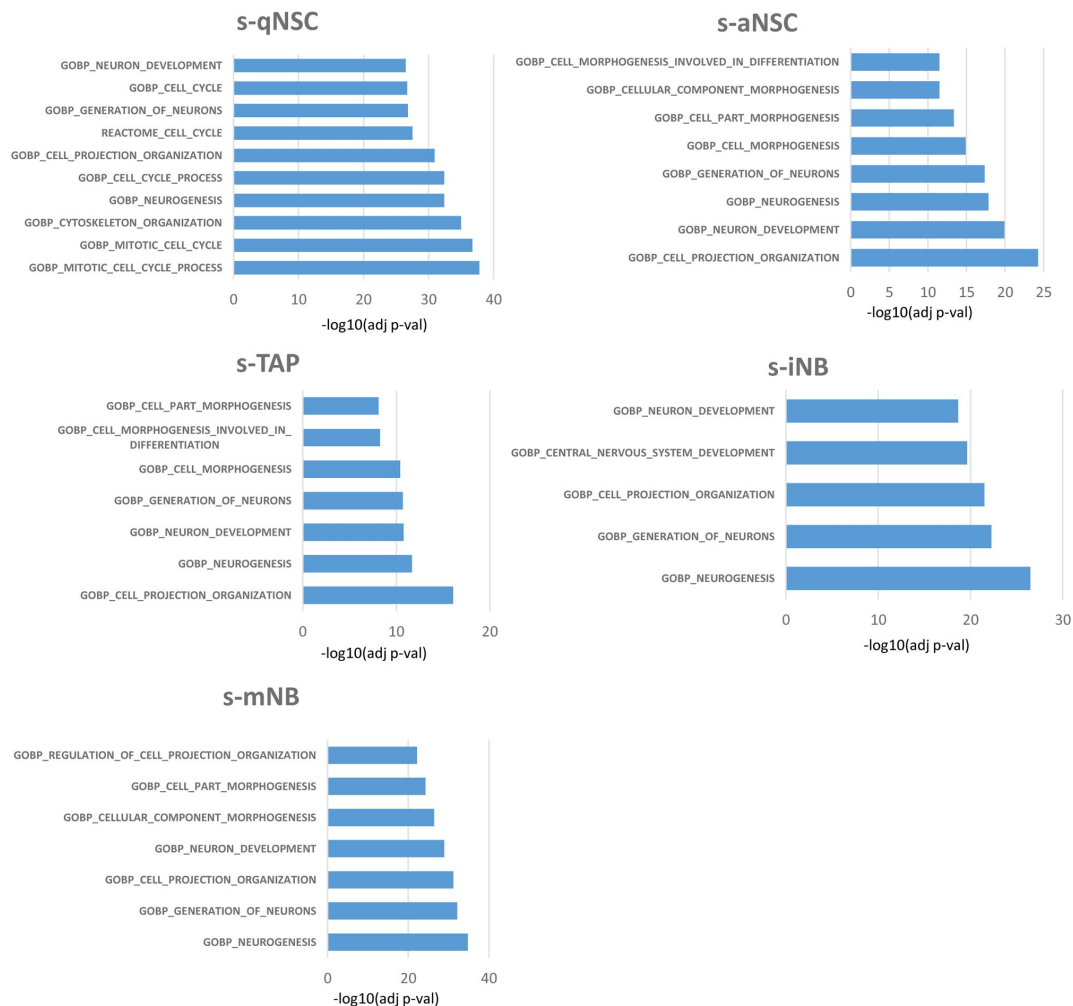


Figure S4

Enrichment of GO terms corresponding to mRNA splicing isoforms in the different types of sorted SVZ cells

Comparisons of individual mRNA splicing isoforms with the Gene Ontology Biological pathway sets using the Gene Set Enrichment Analysis (GSEA) interface. This analysis clearly revealed splicing in genes involved in neuron development and neurogenesis in all SVZ cells. Interestingly, this also showed that s-qNSC logically differed from the other cell types by splicing in genes involved in mitosis and cell cycle. More importantly, GO annotations in differentially spliced isoforms emphasized the close proximity of s-aNSC and s-TAP on one hand and s-iNB and s-mNB on the other hand confirming that s-iNB are closer to s-mNB than s-TAP.

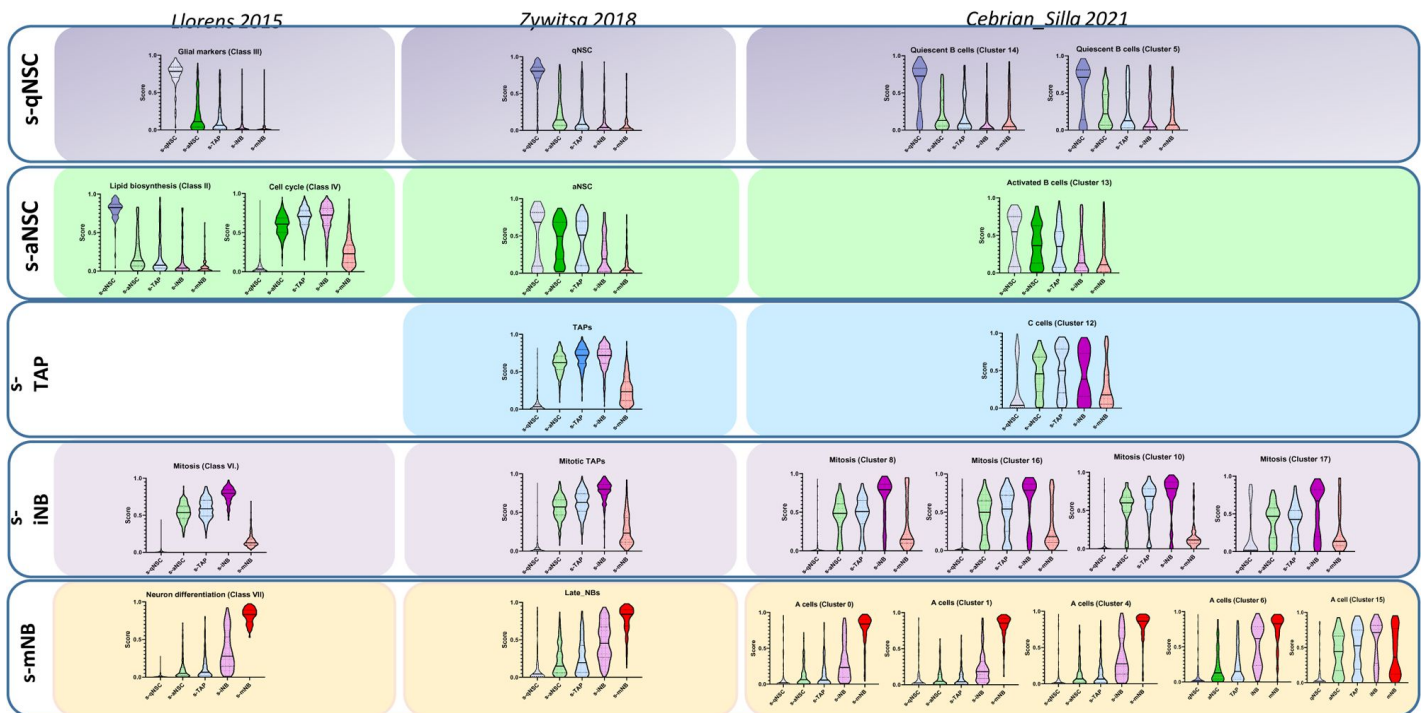


Figure S5

Correspondence between the molecular signatures of sorted neural progenitor populations in our study and available sc-RNA-seq datasets.

Violin diagrams representing the expression of selected clusters from published sc-RNAseq analysis in our Clariom microarray.^{14,17,32} S-qNSC perfectly matched the expression of Glial markers (Class III)³², qNSC¹⁷ and Quiescent B cells (clusters 5 and 14).¹⁴ In contrast, s-aNSC poorly matched with Lipid biosynthesis (Class III) and Cell cycle (Class IV)³², aNSC¹⁷ and Activated B cells (Cluster 13).¹⁴ Similarly, s-TAP poorly matched with TAP¹⁷ and C cells (Cluster 12).¹⁴ S-INB perfectly fit with Mitosis (Class VI), mitotic TAP (mTAP) (Zywitza et al., 2018) and Mitosis (Cluster 8, 10, 16 and 17).¹⁴ s-mNB overlapped genes related to Neuron differentiation (Class VII), Late NBs¹⁷ and A cells (clusters 0, 1 and 4).

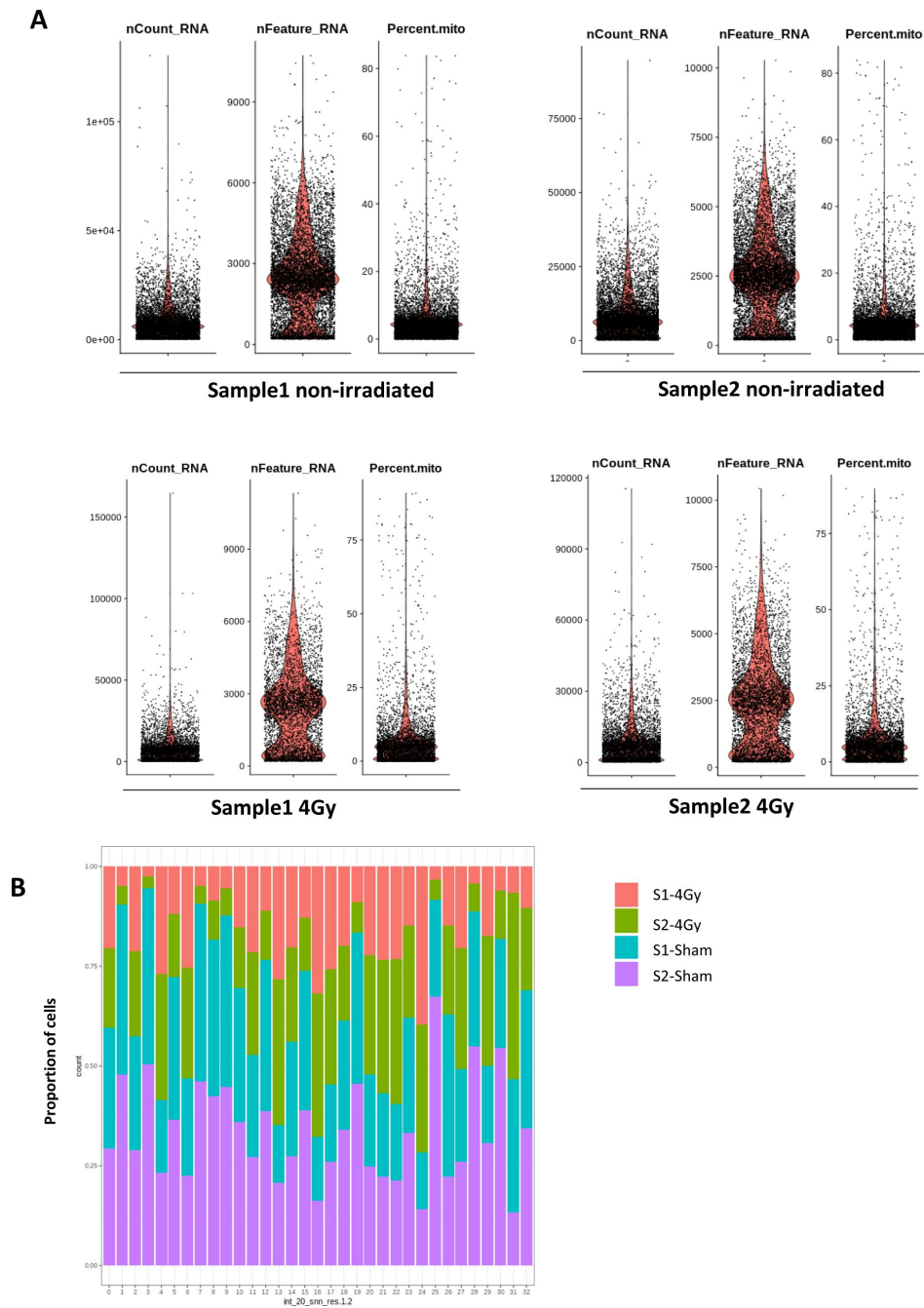


Figure S6

Quality controls of scRNA-seq on whole SVZ tissues

(A) Violin diagrams of the nCount_RNA, the nFeature_RNA and the mitochondrial percentage (Percent.mito) before applying cut-offs adapted to each non-irradiated and irradiated samples. (B) Proportion of cells from the 4 samples in the 33 clusters.

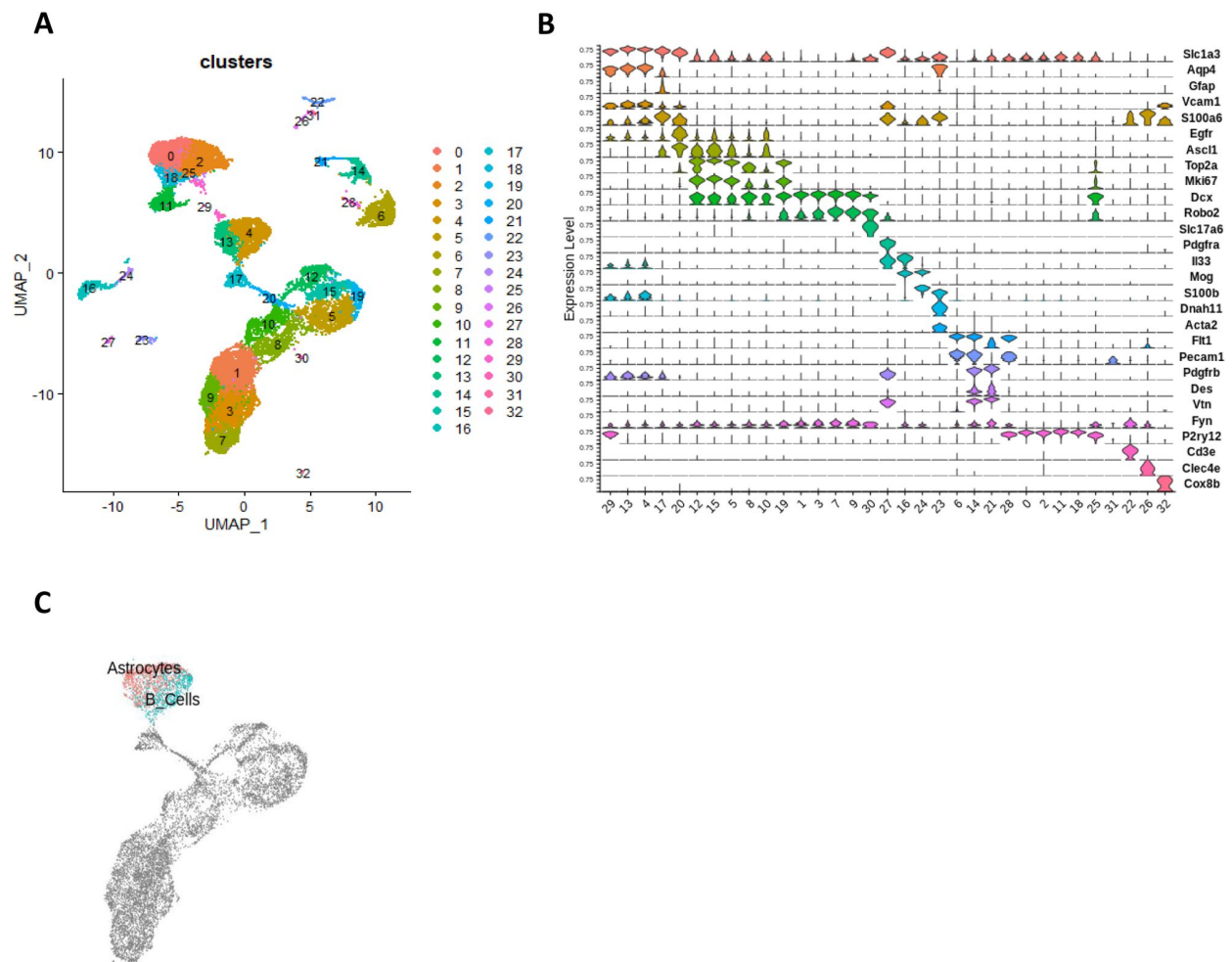


Figure S7

Annotation of sc-RNAseq clusters

(A) UMAP at the resolution 1.2 combining scRNA-seq datasets of unirradiated and 4Gy irradiated cells segregated into 33 clusters. **(B)** Violin diagrams illustrating the expression level of known cell markers in the 33 clusters according to the literature. **(C)** UMAP of top normalized UCellScore Astrocyte- and the B-cell specific genes described by Cebrian-Silla et al ¹⁴ for cells from clusters 4 and 13.

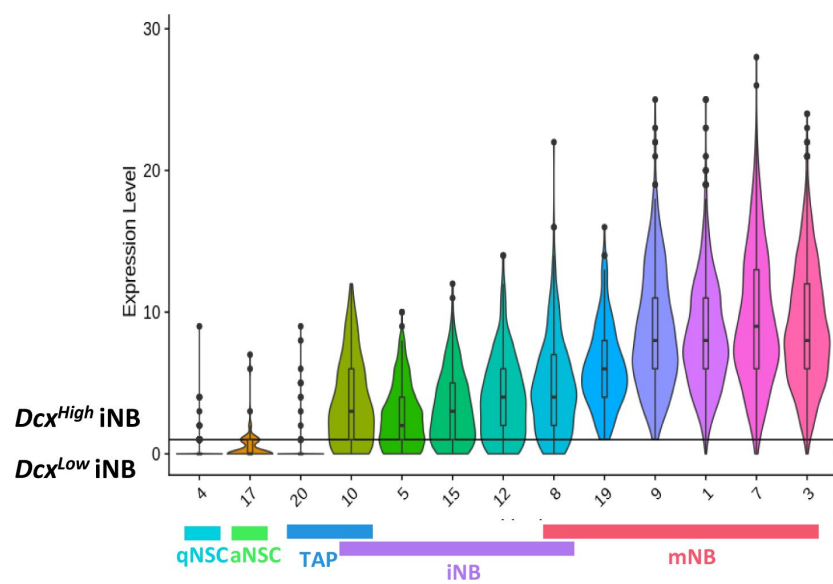


Figure S8

Expression of *Dcx* in scRNA-seq neurogenic clusters

Violin diagram of the expression of *Dcx* in cells from neurogenic clusters illustrating two populations of iNB: iNB *Dc*^{High} and iNB *Dc*^{Low} cells.

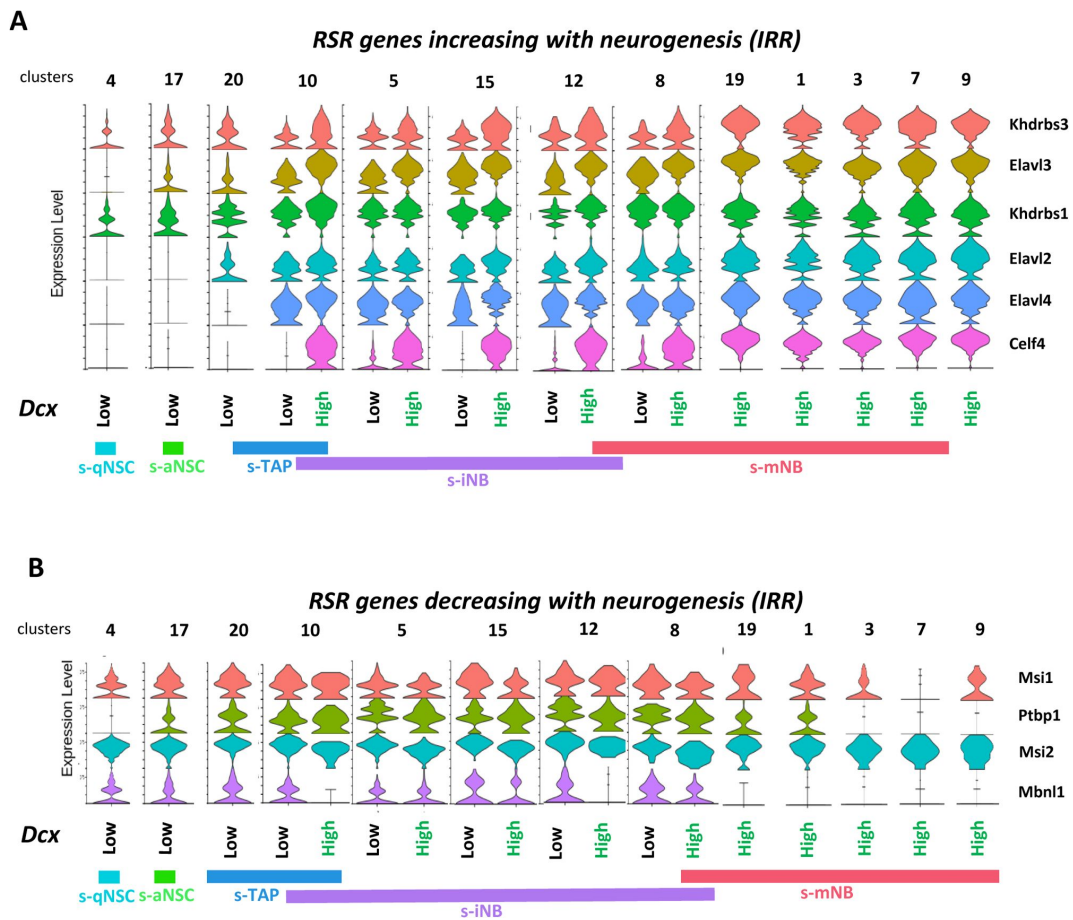


Figure S9

Expression of RSR genes in scRNA-seq neurogenic clusters in irradiated brains

Violin diagrams illustrating the expression of the RSR genes increasing (A) or decreasing (B) with neurogenesis.

		IS			SVZ		CC		OB		
2 weeks after grafting		eGFP+ engrafted cells	Nb eGFP+ cells	%	Nb eGFP+ cells	%	Nb eGFP+ cells	%	Nb eGFP+ cells	%	Total eGFP+ cells
s-INB eGFP+	Animal 1	1.1x10 ⁴	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0
s-INB eGFP+	Animal 2	1.1x10 ⁴	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0
s-INB eGFP+	Animal 3	1.1x10 ⁴	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0

		IS			SVZ		CC		OB		
5 weeks after grafting		eGFP+ engrafted cells	Nb eGFP+ cells	%	Nb eGFP+ cells	%	Nb eGFP+ cells	%	Nb eGFP+ cells	%	Total eGFP+ cells
s-INB eGFP+	Animal 1	1x10 ⁴	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0
s-INB eGFP+	Animal 2	1x10 ⁴	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0
s-INB eGFP+ /eGFP-	Animal 1	2x10 ⁴ / 8.1x10 ⁴	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0
s-INB eGFP+ /eGFP-	Animal 2	2x10 ⁴ / 8.1x10 ⁴	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0
s-INB eGFP+ /eGFP-	Animal 3	2x10 ⁴ / 8.1x10 ⁴	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0
s-mNB eGFP+	Animal 1	2x10 ⁴	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0
s-mNB eGFP+	Animal 2	2x10 ⁴	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0
s-mNB eGFP+	Animal 3	2x10 ⁴	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0

IS: Injection Site
SVZ: Subventricular Zone
CC: Corpus Callosum
OB: Olfactory Bulb

Figure S10

Transplantations of eGFP⁺s-INB, eGFP⁻s-INB and eGFP⁺s-mNB freshly isolated from DCX-CreERT2::CAG-floxed-eGFP mice model

Table recapitulating the immunohistological analyses of the numbers of eGFP⁺ cells in different regions in recipient C57Bl6/J mice brains, 2 and 5 weeks after transplantation.

Supp_data_2: Differentially spliced genes (DSG) unique to s-qNSC, s-aNSC, s-TAP, s-iNB and s-mNB and corresponding Gene Enrichment Analyses (GSEA).

Supp_data_3: Differentially spliced genes in each transition along SVZ neurogenesis and corresponding g:Profiler Gene Ontology analyses.

Supp_data_4: Differentially expressed genes (DEG) among RNA splicing regulators (RSR) genes in each transition along SVZ neurogenesis.

Supp_data_5: Identification of astrocytes and qNSC among clusters 4 and 13 from our scRNAseq dataset using the transcriptomic signatures of Astrocytes and B_cells described by Cebrian et al. 2021.

Supp_data_6: Differentially expressed genes of s-INB Dcx^{High} and s-iNB Dcx^{Low} cells and functional profiling using g:Profiler interface.

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Reviewer #1 (Public Review):

In this study, the authors use prospective sorting and microarray analyses, extended by single-cell RNA sequencing, in the neural stem cell niche of the subventricular zone (SVZ) to identify and refine a series of states along the continuum from quiescent neural stem cells to mature progeny. Of note, changes in the levels and subgroups of RNA splicing regulators are detailed across this continuum. Using in vitro proliferation and differentiation assays, coupled with in vivo engraftment of some prospectively sorted subsets, the authors argue that a stage they define as immature neuroblasts (iNBs) retain proliferative and multilineage differentiation capacity that is not seen in the mature neuroblast population, and is unexpected based on prior models for lineage progression in this system. This iNB stage is accompanied by a change in RNA splicing regulator expression, which is of interest due to the emerging roles for RNA processing and preferential translation within this niche.

The central tension driving the discussion between authors and reviewers, in my view, is what is required to define cells as a "molecularly distinct population" in a lineage. Is it transcript expression, in vitro potential, or more? The authors argue that sorted immature neuroblasts are a defined, separate step in the neurogenic lineage. An alternative possibility is that this population is simply cycling transit-amplifying progenitors that have initiated a transcriptional program associated with neuroblast fates - that these cells are an intermediate point on a continuum between stem cells, transit-amplifying progeny, and commitment to a neuronal (or other) fate. Despite some additions in response to initial reviews, it is still the case that much of the data presented would be equally or more effective in supporting the second model. For example, the differentially spliced gene sets in Figure S4, which are put forward by the authors to support a different molecular identity for immature neuroblasts, show that the terms enriched for immature neuroblasts are largely also found in transit amplifying progenitors (generation of neurons, neurogenesis, cell projection organization, neuron development) and/or mature neuroblasts (cell projection organization, generation of neurons), suggesting that "immature neuroblasts" are transiting between these two states and that one of their most relevant features is that they are still cycling.

These data complement several additional sc-RNAseq studies of this stem cell niche, and use a different, but similar, sorting strategy to isolate and profile subpopulations of stem/progenitor cells and neuroblast progeny. The claim that immature neuroblasts retain multipotency - the ability to generate glia and neurons - is surprising and somewhat controversial given that this has largely not been reported before under homeostatic conditions. Some factors to consider when interpreting these data are that the "immature neuroblast" populations are studied in some experiments using a transcriptional signature and a functional assay, namely the timing of reappearance of these cells after use of agents that kill rapidly dividing cells (in this case, radiation), leading to reconstitution of the lineage by previously quiescent stem cells. In a separate set of experiments, a tamoxifen-inducible labeling system is used in combination with cell-surface markers to prospectively isolate and study the differentiation potential of neuroblast populations that are assumed to be equivalent to those found in transcriptional experiments. It would be of interest in the future to confirm that the exact sorted populations (using CD24/EGFR/DCX-CreERT2::CAG) have the same transcriptional profile as those studied in earlier experiments within the paper and to

confirm the purity of the sorted populations. Finally, while use is made of engraftment of sorted populations to study the differentiation and lineage potential of these immature neuroblasts, a remaining question is the relative abundance of each lineage (neurons/astrocytes/oligodendrocytes) produced by the engrafted cells - is production of glia rare, or common? Could this be due to factors such as alteration of lineage potential due to culture conditions, a disconnect between transcript expression and protein expression, or an incompletely purified starter population?

Overall, this manuscript presents an intriguing possible refinement of models for SVZ neurogenesis, and highlights the role of RNA splicing at specific stages in the lineage. It will be of interest to see if additional groups confirm these findings and whether multiplexed immunostaining, highly multiplexed flow cytometry, or other approaches focused at the proteomic level extend these findings, particularly given recent data in the developing brain that suggest transcript and protein levels are relatively poorly correlated in stem/progenitor populations.

<https://doi.org/10.7554/eLife.87083.2.sa1>

Reviewer #3 (Public Review):

Summary:

Bernou et al. propose the existence of a distinct neuroblast population with increased regenerative and differentiation potential. Their claims are based on the analysis of a sorted population identified as LeX-EGFR+CD24low, which they refer to as "immature NeuroBlasts, iNB". This population is defined by transcriptomics features that have been assessed through bulk microarray studies of sorted cells and single cell RNA sequencing of the whole SVZ-lineage. Analysis of these data sets leads to the identification of these iNBs as cycling cells with a specific expression pattern of RNA splicing machinery components. On these grounds, they propose that RNA splicing plays a key role in neuronal differentiation. Although the authors bring an innovative point to the table, their claims are not fully supported by their results.

Strengths:

Interesting Hypothesis

Weaknesses:

The comparison of their microarray data to published single-cell RNA sequencing datasets (scRNAseq) highlights the cycling nature of the iNB population. Moreover, their own cell cycle analysis on their scRNAseq data attributes G2M/S-phase stages to clusters classified as iNBs, while clusters identified as TAPs are assigned to a restricted G1/S-phase stage. However, it would be expected that TAPs, as cycling progenitors, would go through all cell cycle stages and not just the beginning of it. Thus, authors should consider the possibility that their iNB population entails a major fraction of transit amplifying progenitors (TAP) and a couple neuroblasts, as described in numerous previous studies.

Authors regard the iNB population as neuroblasts due to the capacity of their sorted population to proliferate and differentiate into diverse neural cell types (neurons, oligodendrocytes and astrocytes) in vitro. It cannot be discarded that the sorted population (LeX-EGFR+CD24low) may not be pure and may be composed of a mixture of cells in different stages, including TAPs. Such a mixture of different cell types is unavoidable in sorted populations analyzed as bulk and is precisely one of the issues solved by single cell transcriptomics. Thus, the analysis of single cells resolves transition states at higher resolution and should be preferred over bulk analysis to prevent biases in analysis.

To align the authors' findings with the existing body of literature and earlier characterizations of the SVZ niche, it is advisable to combine their single-cell RNA sequencing data with datasets that have already been published. Such integration will enable precise understanding of the identity of their iNB cells.

On another note, the role of RNA splicing on neurogenesis lacks experimental validation. Unless manipulation of RNA splicing factors is conducted, the key role of this machinery in adult neurogenesis cannot be claimed.

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Author Response

The following is the authors' response to the original reviews.

Reviewer 1

Major points:

R1C1: I appreciate that the data are aligned, in some points, with related studies of this niche. However, it would help the reader to have this alignment explored more extensively in the Discussion as well.

Answer: We acknowledge that the discussion would benefit from additional comparisons to the available datasets. We thus add the following comment after the first paragraph of the discussion: "Previous studies of the different sub-populations of SVZ progenitors were carried out using transcriptomic approaches based on the expression of various more or less specific markers. These approaches have made it possible to identify quiescent and activated neural stem cells as well as mature neuroblasts, but have been faced with the strong influence of the cell cycle on cell clustering. Indeed, neural progenitors in these studies cycling have been gathered in either "mitotic" clusters (Llorens et al. 2015, Zywitzka et al. 2018, Cebrian et al. 2021) or "neural progenitor cells" clusters (Dulken et al. 2017) that had no clear biological significance and hindering identification of subtypes of SVZ cycling progenitors. Our study, combining, for the first time, characterization of FACS-isolated cells and an irradiation-based model of sequential regeneration, allowed to clearly distinguish the molecular profiles of TAP and iNB among cycling progenitors reflecting differences in their in vitro and in vivo respective potentials".

R1C2: The data on multilineage differentiation, both in culture and upon engraftment, would be greatly strengthened by quantification. What is the relative yield of TUJ1/DCX-positive cells versus the other marker combinations? Specifically regarding the multilineage differentiation in vitro - because different media conditions are used to generate each lineage, it may be difficult to determine relative yield. Could a differentiation system that allows production of all 3 lineages be used instead?

If the fraction of non-DCX/TUJ1-labeled progeny is low, particularly in vivo, this might suggest that while multilineage differentiation is possible, it is a much less likely cellular state outcome than production of mature neuroblasts. Some suggested references with examples of the culture conditions, experimental conditions, and discussions highlighted in the public review: Culture conditions that allow simultaneous trilineage differentiation. PMID: 17615304 Influence of culture conditions on potency: similar to issues covered in PMID: 21549325.

Answer: We agree with the reviewer that quantification of a multilineage differentiation in vitro would improve the characterization of the relative potencies of the different SVZ progenitor.

According to PMID: 17615304 and PMID: 21549325, and in agreement with our own experience, the only culture condition that allows neurosphere-derived neural progenitors to differentiate in vitro into the three lineages is the removal of mitogens from the culture medium. However, this does not work on freshly isolated SVZ cells, which remain in an undifferentiated state in this condition.

This is why we chose to use specific differentiation media for each of the 3 lineages as in Figure 1C. It is also for this reason that we performed as many experiments as possible in vivo rather than in vitro as in Figure S2. In the new version, we have added a quantitative analysis of stainings by antibodies against GFAP, CNPase or DCX of GFP-positive cells persisting at IS, where high number of grafted cells were found in Figure S2B. This was performed by using the NIS software measuring eGFP-, GFAP-, CNPase- and DCX-positive areas. The intersection between each marker and eGFP areas was then determined as a percentage of staining (Figure S2C). The results showed that approximately one third of GFP+ cells expressed GFAP or DCX. The quantitative analysis of CNPase expression was complicated by CNPase-positive host cells, but the stronger CNPase staining in eGFP-positive areas clearly revealed the expression of CNPase by a significant proportion of eGFP-positive cells.

R1C3: Additionally, for claims similar to what is currently made in the text, it would be extremely valuable to confirm the purity of the sort for each population - for example by fixing and staining the sorted fraction with additional antibodies that confirm cell identity.

Answer: We have previously shown in Daynac et al. 2013 that s-iNB expressed the neuroblast markers CD24 and DCX, but also markers of neural progenitors such as Mash1, a basic helix-loop-helix transcription factor. As suggested by the reviewer, we have further investigated the expression of other markers of neural progenitors by sorted cells. The results showed that the proportion of DLX2+ cells a marker of proliferating progenitors (Doetsch et al. 2002) was very high in aNSC/TAP (98%) and progressively decreased in iNB (82%) and mNB (25%). Similarly, the expression of the transcription factor SOX2 that plays an essential role in the maintenance of neural progenitors (PMID: 25126380) accounted for 78% of aNSC/TAP, 70% of iNB and 17% of mNB.

Altogether, these new data confirmed the identity of the different cell populations and particularly that of iNB. They are commented at the beginning of the Results and shown in Figure S1.

R1C4: Line 125: GFAP alone doesn't necessarily indicate a "conversion to NSCs" - this conclusion could be greatly strengthened by inclusion of more markers, particularly at the protein level, or cyto-architectural studies.

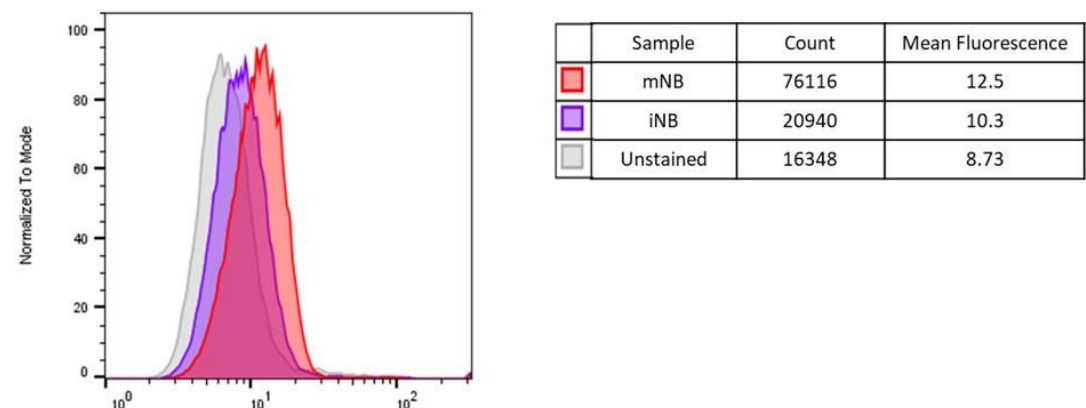
Answer: We agree with the reviewer that GFAP expression alone is not sufficient to evidence the presence of NSC in the SVZ. We have thus modified the text accordingly: "Importantly, eGFP+ cells were present in the SVZ of all the animals transplanted with eGFP+s-iNB and eGFP+s-NSC/TAP (Fig. 1Db, Fig. 1Dc), some of them expressing GFAP indicating the generation of astrocytes, and therefore possibly NSC".

R1C5: Could these cellular states be reflective of preferential translation of DCX? It would be very helpful to see the flow cytometry sort data for iNBs / mNBs used in Figure 6, particularly if these cells were also fixed and stained directly for DCX protein.

Answer: As suggested by the reviewer, freshly FAC-sorted iNB and mNB were fixed and labelled with an anti-DCX monoclonal antibody after permeabilization. As shown in the figure below, we found a higher level of DCX expression in mNB than in iNB. Therefore, this result tends to indicate that the proliferation capacity is somehow related to the level of DCX expression. However, because of the relatively low importance of this result, we decided not to include them in the manuscript.

Author response image 1.

Modal histogram representation of DCX expression level in unstained, iNB and mNB cells determined by flow cytometry (FlowJo).



<R1C6: Figure S8 is all zeroes, showing the GFP+Dcxhigh NBs do not retain proliferative capacity. But we don't get a direct experimental comparison to EGFPnegative/lowDcxlow iNB engraftment, which would strengthen the conclusions of the paper.

Answer: Unfortunately, there is no method available to analyse the eGFPnegative/lowDcxlow iNB engraftment: by definition, these cells do not express eGFP and the use of a tracker is not appropriate for long periods of time — and thus a high number of cell divisions — after engraftment. However, to us, this control is not needed to conclude that GFP+Dcxhigh iNB have no (or at least a lower) stem cell potential in vivo considering that we have shown in Figure 1 and Table 1 that the whole iNB population is able to generate the different types of neural cells.

R1C7: Transplant data in Table 1 - a relatively small proportion of transplant derived cells are in OB, etc. Given that A cells are thought to cycle at least once in vivo, is this expected?

Answer: The reviewer is right considering that a relatively small proportion of transplant derived cells were found in the OB. However, we should consider that we used immunocompetent mice as receivers, which could have significantly reduced the engraftment efficiency, and the migration of engrafted cells outside the injection site.

R1C8: A caveat is that there is not much functional testing of the proposed model, especially for the interconversion of iNB states suggested by the diagram in Figure 7. The text is relatively restrained in proposing this model, so it is reasonable to keep - but perhaps should be noted that this part of the model will need additional testing.

Answer: Data presented in Figure 6 clearly suggest that Dcxhigh iNB have similar in vitro potential than Dcxlow iNB, whereas they don't have such potential in vivo (Figure S10). This suggests that, providing they are in appropriate conditions, Dcxhigh iNB could reacquire stem/progenitor properties. However, we agree that this hypothesis requires further investigation. Therefore, as suggested by the reviewer, we have added in the Figure 7 legend: "Possible interconversion of iNB states would require further experimental confirmation."

Additional minor points:

R1C9: Introduction: the SVZ is described as "the lateral wall" - however, several works in the mouse have also examined the medial wall and callosal roof, as cited later in the intro. Suggest rephrasing the second sentence (line 48) and later sentence (line 66) to clarify that "the SVZ" encompasses all of these subregions, they are not necessarily separate niches. Answer: As indicated by the reviewer, the SVZ encompasses distinct subdomains, with NSCs having a regional identity based on their location in the lateral or septal wall of the ventricle and generating different types of neuronal and glial progeny (PMID:34259628.). To address the reviewer concern about possible confusion and clearly indicate that SVZ encompass several subdomains, we have modified the sentence line 66 as follows: "Since then, the single cell RNA-sequencing has revolutionized the field and has made it possible to precisely elucidate the transcriptome of SVZ cells present in the LW and in the septal wall which also harbors NSC niches".

However, we did not modify the line 48, since in this sentence we just indicate that the largest neurogenic niche in the adult brain reside in the LW of the SVZ.

R1C10: Line 77: "exposure" not "exposition"

Answer: The error has been corrected in the revised manuscript.

R1C11: As noted in the Public Review - the use of the term "D1/D2" cells seems likely to confuse readers who are also versed in dentate gyrus neurogenesis. Recommend removing this term from the manuscript.

Answer: We agree that the D1/D2 terminology could bring confusion, D cells referring to Tanycytes in the hypothalamus. We now refer to iNB1 for DcxLow iNB and iNB2 for DcxHigh iNB in the revised manuscript.

Reviewer 2

Major comments:

Lack of rigor

R2C1: There is a lack of appropriate normalization controls for the microarray data. As there is a decreased level of transcription in quiescent NSCs, there needs to be a cell number control (spike-ins based on cell numbers). Without this normalization, the readout can be greatly skewed.

Answer: We agree that qNSC are marked by a decreased level of transcription due to quiescence. To overcome this problem in the Clariom assays, we thus chose to calibrate each population, with a fixed amount of cRNA and cDNA using Hela cells as internal control. We totally agree that this method is not optimal but it appears to be efficient in the end. Indeed, it should be noticed that it has been adopted, thus with the same rigor, in other microarray studies published in the field (PMID: 24811379) and also on skeletal muscle cells (PMID: 29273087). Moreover, interestingly the transcriptomic signature of qNSC matches perfectly

with those from other studies and particularly to those of related clusters in single cell experiments (including ours, Figure S5). This is probably linked to the fact that more importantly that the number of cells, the main characteristic of these cells is the lack of expression of genes involved in cell proliferation and metabolism. Whatever so, these data confirming previously published are not the main information of our manuscript, which is mainly dedicated to the characterization of proliferating cells, which is not impaired by our choices of normalization.

R2C2: The absolute segregation of clusters in the single-cell analysis is currently entirely in agreement with the cell cycle stage. This suggests that in the author's analysis, the clustering in 3F is entirely shaped by the cell cycle, making that the defining characteristic of the author's definitions for their cell types. Has an analysis been done that regresses out cell cycle-associated genes to see if there are clusters for different cell states/types that are identified in the absence of cell cycle stage being the defining factor? (Barron and Li, 2016). For example, just as you would see a difference in cluster if you are a quiescent or activated NSC as compared to a neuroblast for example, even without the contribution of cell cycle. These are different cell types.

Answer: We agree that cell cycle regression would theoretically allow for further discrimination between cycling cells along successive neurogenic stages. We have already performed regression using several methods, including regressing using S- and G2/M-score regression as indicated in the Seurat workflow, removing cell cycle-related PCs from UMAP calculation as used in the Cebrian-Sylla study, and using alternative gene sets such as the ones provided by the tricycle method (PMID: 35101061). These regression methods have all been used on our datasets, the original Cebrian-Sylla datasets and a combination of our datasets with the Cebrian-Sylla original datasets to increase cell number and clustering resolution. However, none of these methods modified the clustering of cycling cells.

In fact, the strong influence of the cell cycle over clustering highlights the relevance of our depletion/replenishment approaches to decipher the molecular changes masked by the cell cycle, as discussed below.

R2C3: The use of the DCX-CreERT2 line is a lineage tracing line. Once DCX is expressed, Cre recombines the DNA to allow for fluorescence. It is binary, on or off associated with DCX expression. And once on, it is always on, whether the cell is currently expressing DCX or not. As the authors had previously described a DCX^{low} condition, the eGFP⁻ cells would not reflect DCX^{low}, but no DCX at all. And the eGFP⁺ cells may not be currently expressing DCX anymore. The authors should have used a system where the DCX promoter itself drives fluorescence.

Answer: We took advantage of the DCX-CreERT2 line to demonstrate that some neural cells that have recently acquired DCX expression (i.e. eGFP⁺ iNB) could keep (or recover) the potential of neural progenitors in vitro. Of course, some of these GFP⁺ cells could have stopped to express DCX. This is probably the case when they differentiate into astrocytes and oligodendrocytes in vitro as shown in Figure 6.

Whatever so, the use of the Dcx promoter as a direct driver of eGFP fluorescence would have totally impeded our capacity to demonstrate such changes in cell fate in vivo because of the impossibility to track oligodendrocytes or astrocytes derived from iNB because of the loss of Dcx expression.

R2C4: The lack of analysis of images (differentiation, for example) limits the conclusions of the in-vitro data, and the images with unclear staining, limit the conclusions of the in-vivo experiments.

Answer: This comment is similar to that of R1C2. We have now added a quantification in Figure S2.

R2C5: The cited difference in splicing differences in cell types was interesting (though did not show up in the transcriptome enrichment analyses Fig S2) and would be something to further pursue, however, this was a very limited analysis. There was no further study of these splicing mediators beyond single-cell data.

Answer: We now show enrichments of GO terms corresponding to mRNA splicing isoforms in the different types of sorted SVZ cells (Figure S4). This analysis clearly revealed that spliced genes in SVZ cells are mainly involved in neuron development and neurogenesis. Interestingly this also showed that qNSC logically differed from the other cell types by splicing concerning genes involved in mitosis and cell cycle, consistently with their quiescent state. More importantly, GO annotations of differentially spliced isoforms further confirmed that s-TAP and s-iNB have distinct features. We agree with the reviewer that further analysis of splicing mediators would be very important for understanding molecular changes involved in neurogenesis. However, we think that it is largely beyond the scope of this study.

R2C6: Fig 1C - Show values, not just pictures. You may need to shift your current differentiation paradigm to do so by removing growth factors instead of unique differentiation conditions.

Answer: See the answer to R1C2.

R2C7: Fig S1A - Stainings for GFAP and DCX are not clear. It is very hard to distinguish which cells are associated with these signals.

Answer: This figure (now Figure S2A) shows an eGFP+iNB cell (white arrow) that has reached the rostral migratory stream and expressed DCX (inset a3), but not GFAP (inset a2). This is now indicated in the figure legend. We have also moved the arrow for more clarity.

R2C8: Fig S1B2 - There is red staining everywhere, so it is very hard to see a specific CNPase signal.

Answer: We have added a new figure (Fig S2B) distinguishing eGFP+CNPase+ cells (yellow arrows) from eGFP+CNPase- cells (white arrow).

R2C9: Line 174 - It's the mRNA that you are detecting is being downregulated - be more specific as you are not showing protein downregulation.

Answer: We specified, "encoding" a major splicing repressor in the Line 174 text to refer to the mRNA: "Interestingly, Ptpb1, encoding a major splicing repressor".

R2C10: Line 189 - text in this line have some clusters not shown in the figure - (clusters 6 and 15, DCX+ Ki67+ neuroblasts) - which would be an important thing to visualize. As is shown now, the authors are only showing that iNBs are similar to mitotic TAPs.

Answer: Clusters 6 and 15 have been added to Figure S5.

R2C11: Fig 3D-E - Why is cluster 17 called aNSCs (3E) when it has the highest GFAP (Fig 3D). Typically, the highest GFAP cells are qNSCs or astrocytes, not aNSCs.

Answer: We previously reported that the level of gfap mRNA expression in neural stem cells (quiescent and activated) did not exactly reflect the amount of protein in these cells. This is the reason why we also used the Slc1a3 marker (Glast), which is highly expressed both at the RNA and protein levels in quiescent NSCs (Daynac et al. 2013).

R2C12: Line 216 - You said in line 216 cluster 13 were astrocytes, then you said in line 227 that cluster 13 was s-qNSC. Which is it?

Answer: This is due to the fact that we performed two distinct analyses.

In the first one (line 216), cells were scored based on datasets provided by Cebrian et al. with one dataset containing genes enriched in astrocytes, and another one, genes enriched in quiescent B-cells. Therefore, cluster 13 was shown to contain 73% cells expressing astrocyte markers, whereas cluster 4 gathered cells expressing both qNSC (B-cells, 48%) and astrocyte (52%) genes.

In the second one (line 227), cells were scored using our transcriptomic signatures of FAC-sorted SVZ cells, which do not include differentiated astrocytes. We demonstrated that the cluster 13 cells only expressed s-qNSC genes.

R2C13: Line 214 - While other clusters were all named in lines 214-221 that were then further discussed in lines 227-230, clusters 15 and 19 were not. You associate both of those clusters with s-iNB - what was it associated with in the above section?

Answer: Lines 219-221 have been reworded as follows: Clusters 10, 5, 15, 12, and 8 were defined as cycling progenitors based on the expression of proliferative markers such as Top2a, Mki67, Ascl1. Clusters 1, 3, 7 and 9 were identified as mNB due to the loss of Mki67, Top2a and Ascl1 expressions and the expression of Robo2 and Dcx. Cluster 19 that have lost Ascl1 but still expressing Top2a and Mki67 together with Robo2 and Dcx appears at the transition between iNB and mNB.

R2C14: Fig 3I-J - 5 days after irradiation, I would like to see from tissue slices how many cells are dividing compared to 1day post-irradiation and controls. In other paradigms, such as temozolomide experiments (Kalamakis et al), by 5 days we should see less cells in quiescence and more of those quiescent cells exiting quiescence into the cell cycle. Why would there be more cells in quiescence in the irradiated brain? Even if they are radiation resistant, the base number should be comparative between controls and irradiated, which is not what you show in Fig 3I-J. And R2C14)

Line 234-235 - the text says normalized to numbers of qNSCs which is supposed to be the same (which I agree should be the same). However, your graph in 3I and J shows more qNSCs in irradiated conditions, which would influence greatly and is currently hard to interpret.

Answer: As stated by the reviewer, there is no increase in the absolute number of quiescent cells in the irradiated SVZ. The reconstitution of SVZ cell populations after 4Gy irradiation has already been studied by our group (Daynac et al. 2013, see Fig. 3F), showing that s-iNB and s-mNB are still under-represented after 5 days, while qNSC are in similar numbers as in unirradiated SVZ. Therefore, this led to an over-representation of quiescent cells and early SVZ progenitors in Figure 3J as compared in Figure 3I.

R2C15: Fig 6A - the authors show a significant difference in neurospheres between eGFP- (DCX-) and eGFP+ (DCX+) iNBs - as would be expected as DCX suggests a further commitment towards neurogenic fates, yet your population doubling is the same.

Answer: To determine the population doublings, the medium was changed and cells numbered every 7 days. This condition masked the differences between two cell populations reaching the plateau phase at different time, explaining why eGFP-iNB and eGFP+iNB could not be clearly distinguished by this technique.

R2C16: Fig 6C - Differentiation data (in-vitro) should be quantified in 6C, just as was mentioned for 1C. These values should be done for both of the populations (eGFP-iNB, and eGFP+iNB) and not just compared to the previous pictures which were on total iNB. Again, numbers are required, not just picture examples.

Answer: Quantitative data have been given in Figure 6D showing that approximately 60-80% of cells eGFP+iNB are able to differentiate in either neurons, oligodendrocytes or astrocytes. We did not analyze the differentiation of eGFP-iNB since it would not add any supplementary information.

R2C17: Fig S8 - The authors did not show if the lack of engraftment of eGFP+ cells is due to the transplant (previously you showed only 2/3 worked in a similar paradigm). It would be helpful if the authors would have some means to visualize the DCX low cells to confirm they worked as before in the transplantation (another color? Another type of mouse (Thy1 antigen differences)?) Answer: Unfortunately, the Thy1 antigen has not been documented in mouse subventricular zone progenitors, but only in neurons (PMID: 10813783). Thy1 antigen has also been described in bipotent glial progenitor cell (GCP) from the developing human brain giving rise to oligodendrocytes (PMID: 36931245).

As shown, in Figure S10 we have performed 5 grafts with s-iNB eGFP+ cells, 2 alone and 3 mixed with eGFP- cells and never found any eGFP+ cells 5 weeks after grafting. Moreover, we did not find any eGFP+ cells in the brains of 3 other animals 2 weeks after grafting with s-iNB eGFP+ cells (These data have been added to Figure S10). As compared to the results described in Figure 1 this clearly shows that iNB DCX^{high} are not able to generate persistent cells in the grafted brains similarly as mNB.

R2C18: Fig S8 - Why were there no eGFP cells even at the injection site? DCX expression promotes migration, indeed DCX expression becomes very high in cells in the SVZ as they begin to exit to go to the migratory stream. If one didn't see migration, one would expect you would still have survival. Currently, the authors show no cells at 5 weeks, however, they would need to show earlier timepoints as well to determine what is happening with these cells. It is possible these GFP+ cells are not even expressing DCX anymore (see above).

Answer: As stated above, we did not find any GFP+ cells in the brains of 3 other animals 2 weeks after grafting with s-iNB eGFP+ cells (see Figure S10).

R2C19: Line 320 - the authors suggest a subpopulation of NEURONS continues to divide and cite 2 works from the 1990s showing proliferating SVZ cells can differentiate. Our knowledge of this system has come dramatically forward since the 1990s as well as technologically, and to date, neurons have not been shown to divide.

Answer: We apologize for this lack of clarity, as we agree that neurons correspond to differentiated non-cycling cells, but we used the terminology used in these articles. The incorrect part of the sentence Line 320 has thus been deleted from the text.

R2C20: Fig 7 - The whole figure is based on changing levels of RSR genes which were not confirmed in any way to be involved in any of these stages, only descriptively in single-

| cell analyses.

Answer: As stated above, in our opinion, further characterization of the involvement of RSR genes in neurogenesis is largely beyond the scope of our manuscript. Nevertheless, we think that the role of RSR genes in neurogenesis is an important question that should be addressed in further studies.

| *Overstatement of findings*

R2C21: Fig 1 - Authors did not compare all cell types in each condition but made overstatements about their relationships to each other between graphs. There should also be separate graphs showing all cell types at 4% and a separate one at 20%.

Answer: In the revised version, Figure 1 shows the graph comparing all cell types at 4%O₂ and a separate one at 20% as requested by the reviewer. The graphs clearly shows that 4%O₂ promotes iNB proliferation compared to the 20% condition.

R2C22: Fig 1D-b2 - Why does DCX look nuclear? One can't say they are only NSCs if they are GFAP as astrocytes also express GFAP. The authors would need another marker to separate those populations. In the text, the authors say expressing GFAP (line 124) which means NSC, but then in line 127 expressing GFAP means astrocytes - which further shows you need additional markers to validate those 2 different cell types. Answer: DCX nuclear translocation has been shown to improve cellular proliferation (PMID:32050972).

As indicated in R1C4. The text has been modified as follows: “Importantly, eGFP⁺ cells were present in the SVZ of all the animals transplanted with s-iNB eGFP⁺ and s-NSC/TAP eGFP⁺ (Fig. 1Db, 1Dc), some of them expressing GFAP indicating the generation of astrocytes, and therefore possibly NSC”.

| *R2C23: Fig S2 - The transcriptome signature for s-iNBs is very similar to s-TAP, basically suggesting the iNBs are further along in cell cycle.*

Answer: This is now the Figure S3. Functional enrichment analysis of individual transcriptome signatures revealed that both s-TAP and s-iNB are enriched in genes related to the cell cycle although with different GO terms enrichments. Indeed, s-TAP are enriched in genes related to G1, G1/S and S phase (but with low -log₁₀ adjusted p-values) and s-iNB with genes related to cell cycle mitosis and M phase (with high -log₁₀ adjusted p-values).

We have previously shown that around 33 % s-iNB have DNA content>2N, versus around 26% of s-TAP and s- aNSC (Daynac et al. 2013), which is in accordance with GO terms enrichments. However, these data have also shown that most s-iNB and s-TAP are in G1, indicating that siNB are not just further along mitosis than TAP.

Moreover, our transcriptomic data clearly show that s-iNB are distinct from s-TAP: 1) according to principal component analyses (Figure 2B et C), the whole transcriptome of s-TAP is closer to that of s-aNSCs than to that of s-iNB (10% variations in PCA2), 2) the heatmap in Figure 2D shows that they have different RSR genes expression profiles, 3) the new Figure S4 shows that GO annotations of differentially spliced isoforms further confirmed that s-TAP and s-iNB have distinct features, and 5) Figure S5 shows that s-iNB expressed genes associated to either TAP or NB that have been described in previous studies, whereas s-TAP did not express genes associated to NB, but look closer to aNSC. Finally, scRNAseq cell clusters related to s-iNB are distinct from the cluster related to s-TAP as shown 1) in Figure 3D and 2) in Figure 4.

| *R2C24: Fig 3 - The lack of information about timepoint 0 after irradiation, and when proliferation and cell cycle entry begins again following irradiation, limits our*

interpretation of the single-cell irradiated data.

Answer: We have previously reported the relative abundance of each SVZ neural progenitors in the young adult mouse brain in several papers. Particularly, we based our interpretation on our SVZ irradiation model reported in Daynac et al. 2013 demonstrating a radio resistance of qNSC re-entering into the cell cycle as early as 2 days after 4Gy irradiation successively regenerating aNSC, TAP then iNB and mNB.

R2C25: Fig S3 - These results effectively show that the s-aNSCs and s-TAPs are actually less specific when compared to that same identity in other studies, and that the iNBs are most similar to mitotic TAPs. This supports what was mentioned above, which is that the transcriptional signatures are very similar between the s-TAPs and i-NBs, showing these are not a unique cell state, but just a bit further along mitosis within the TAP cell state.

Answer: This is now the Figure S5. In this figure, we show that s-iNB expressed genes associated to either TAP or NB that have been described in previous studies, whereas s-TAP did not express genes associated to NB, but look like closer to aNSC. As indicated above in R2C23, s-iNB are not just a bit further along mitosis within the TAP cell state. Indeed, we give several data showing that s-iNB and s-TAP have different transcriptomic profiles.

R2C26: Fig 4B - The focus on Ptpb1 as being associated with the iNB cluster border to mNB is expected as all previous studies of Ptpb1 have focused on its role in the progression of other cell types through the cell cycle, its control of cell cycle regulators, and a cell cycle mRNA regulon (Monzon-Casanova et al, 2018, 2019, 2020). This further supports these analyses are specifically defined by cell cycle stages.

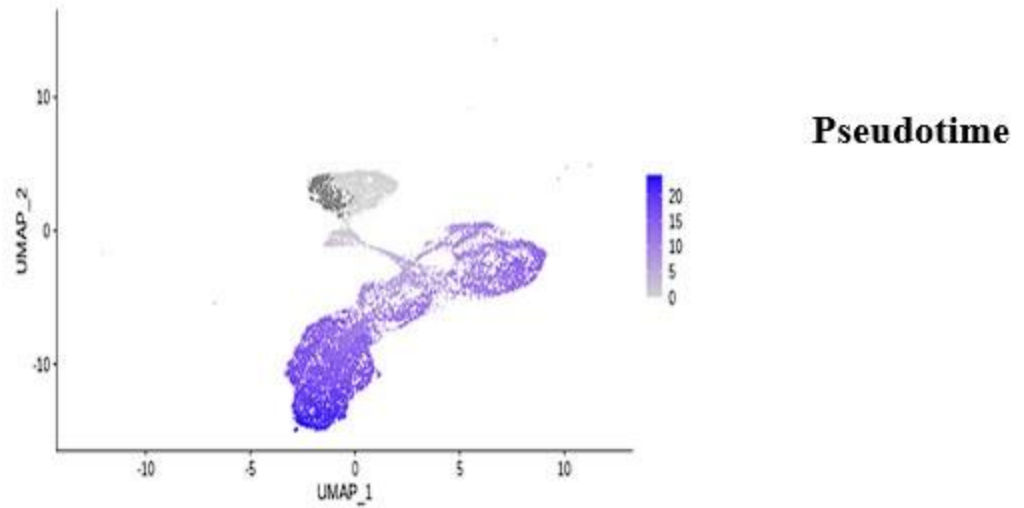
Answer: We totally agree that Ptpb1 expression distinguishes cycling cells from postmitotic neuroblasts in accordance with previously published paper, and that based on this unique gene we cannot find any differences between cycling cells ie. aNSC, TAP and iNB. However, as shown in the manuscript and stated above (R2C23 and 25), these cells can be distinguished by their respective expression of many other genes, including other RSR genes.

R2C27: Line 281-282 is an overstatement - the authors suggest that this is a new type of cycling neural progenitor - when all studies point to it being the end of mitosis TAPs as they go on their way to mNBs. This clearly shows a trajectory and not a defined, binary cell type.

Answer: We agree with this statement that the use of the word "type" was misleading, and changed it to "stage" to better reflect that s-iNB are a distinct stage along the differentiation process according to our pseudotime cell-trajectory analysis.

Author response image 2.

Pseudotime analysis using Monocle 3 (excluding the cluster 13 corresponding to astrocytes and starting from s-qNSC) revealed two branches starting from s-TAP, one towards cell cycle the other towards neuronal differentiation.



minor comments:

R2C28: Fig 3D - For ease, please define what you called the clusters in 3D - not just cluster numbers

Answer: We chose not to call the clusters in 3D because their identification (Group names) is based on data presented after in Figures 3E, F and G.

R2C29: Fig 3E-F - Show astrocytes by text in 3E and F

Answer: As discussed above, astrocytes cannot be shown in these figures because they are based on our signatures which did not include astrocyte signature.